A Paired-Phase Construction for Ordinary Binary Polynomial Multiplication

Árpád Csizmadia
arpad.csizmadia.public@gmail.com

2 November 2026
Version 0.2.1

Abstract

We construct a deterministic algorithm for multiplying two ordinary degree-$< n$ polynomials over $\mathbb{F}_2$ in $O(n \log^{1-1/1302}(n+2))$ bit time on a machine with a fixed finite alphabet and a fixed finite number of sequential tapes. The construction uses a finite family of paired Fourier phase operators, an integer composition of jobs of different ranks, and an interchange algorithm obtained from a linear-time-per-address-bit seed. A tensor algorithm on mixed-radix address arrays then supplies the kernels needed by a cyclotomic Chinese remainder reduction. The bound includes input and output conversion, public setup, coefficient arithmetic, tape movement, workspace restoration and all $2n-1$ ordinary product coefficients. The technical appendices give the finite word, recursive schedules and cost substitution. Exact finite certificates accompany the finite identities and rank inequality.

AI assistance and formal verification. Substantial generative AI assistance supported mathematical exploration, algorithm and proof development, coding, writing, literature review and internal source review. OpenAI Codex was the preparation harness. Exact primary model snapshots and the complete earlier model inventory were not recorded; the accompanying provenance files identify the recorded review models. The named author directs the project and makes publication decisions. Independent human mathematical review and a complete formal proof of the main theorem remain outstanding.

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Background and contribution

Fast polynomial multiplication over finite fields builds on a substantial body of work in arithmetic complexity and structured Fourier transforms. Harvey, van der Hoeven and Lecerf [11] obtained the fixed-prime bound $O(n \log n\, 8^{\log^* n})$; Harvey and van der Hoeven [1] subsequently obtained $O(n \log n\, 4^{\log^* n})$ and developed the cyclotomic coefficient-ring methods used here. Their later theorem gives $O(n \log n)$ over each fixed finite field under an explicit Linnik hypothesis [12, Theorem 1.2]. Van der Hoeven and Larrieu's Frobenius FFT [13] gives a complementary approach that exploits finite-field symmetries, with applications to binary polynomial multiplication. These results provide important arithmetic and transform foundations for the present work.

The earlier controlled-CRT manuscript [3] states a saving $\kappa = 2^{-38}$ in the logarithmic exponent for ordinary binary polynomial multiplication. The present revision states $\kappa = 1/1302$ for the same coefficient input/output problem and deterministic fixed-tape model. It retains the cyclotomic host and controlled-CRT outer framework while developing the paired-phase construction, integer mixed-rank compiler and movement bootstrap used for the improved bound. The finite-network and local Fourier methods of OpenAI [4, 5], Sussman's mixed-job construction [6], and the finite precompilation and five-stage constructions of Colkitt and contributors [7, 8] supply essential antecedents. The contribution studied here is their adaptation and composition with fully charged ordinary binary polynomial input and output. The cited final integer-multiplication and complex-arithmetic bounds are not premises of this theorem.

Relationship to the companion papers. The Pascal/Taylor companion [14] gives a separate binary $O(n \log(n+2))$ construction. Its complete multiplier is not a subroutine of this proof. The general finite-field companion [15] combines the earlier controlled-CRT binary result with an independent odd-characteristic construction. The present proof does not invoke that combined conclusion, so this dependency is not circular. This revision concerns $\mathbb{F}_2$; the other selected field statements retain their own bounds, sources and evidence. The improvement is an asymptotic claim, with no practical speedup or exhaustive priority determination asserted.

1 Statement, model and proof dependencies

Input consists of two ascending coefficient lists of length n over $\mathbb{F}_2$. Output consists of all $2n-1$ ordinary coefficients, including zeros. One deterministic machine has a fixed finite alphabet and a fixed finite number of one-dimensional sequential tapes. Heads move one cell per step. Shape descriptors are ordinary binary integers; address coordinates are public positions and never input-payload controls. The two input tapes contain the end-marked bit strings $a_0 \cdots a_{n-1}$ and $b_0 \cdots b_{n-1}$, with heads at their first symbols. All work tapes and the output tape are initially blank. At termination the output tape contains the end-marked string $c_0 \cdots c_{2n-2}$, where $c_i = [X^i](\sum_j a_j X^j)(\sum_k b_k X^k)$; its head is returned to the first symbol. Inputs may be preserved by copying them once. Counting, copying, final head returns and erasure of marked work intervals are included in the time bound.

Theorem 1. *There is such a machine multiplying ordinary degree-$<n$ polynomials in $\mathbb{F}_2[X]$, for every $n \geq 1$, in*

$$O\Big(n \log^{1-1/1302}(n+2)\Big)$$

bit time. The constants and finite direct-branch cutoff are effective and fixed independently of n.

The time bound counts all transitions of the machine. A subroutine's volume is the number of bits in its current input array, including spectator coordinates, retained padding and coefficient lanes. Traversal of suspended data is charged to the caller that performs it.

The cyclotomic host and principal-root construction is the deterministic construction of Harvey and van der Hoeven [1, Propositions 3.4,3.8]. Short auxiliary products use classical

fixed-tape integer multiplication [2]; they do not call the algorithm being constructed. The outer reduction develops the algorithms of [3], with its costs derived from the kernels proved here in Appendix G. The finite-network and local Fourier methods have antecedents in [4, 5]; mixed-rank composition and finite precompilation are developed in [6, 7, 8]. These works are used through the constructions stated below, rather than through a transfer of their final integer or complex-arithmetic bounds.

The proof is organized into three steps. First a finite all-input word and an integer composition argument produce a uniform rank-N template. Its rational address interpretation gives a fast interchange procedure. Its characteristic-two interpretation, implemented on every active and idle address pattern, gives the native tensor kernels. Finally a controlled Chinese remainder and Fourier reduction returns the ordinary coefficient list. Appendices A–G provide the full constructions and their cost interfaces.

Terminology. A native array is an ordinary lexicographic stream indexed by a Cartesian product of integer ranges. A productive port has arbitrary independent input and receives its specified operator. A paid query is a recursive kernel invocation charged on its complete input stream. Owned arrays are auxiliary streams initialized within the program; their queries are charged, and they receive no independent productive credit. A dirty address coordinate may have any initial value and must be restored at the stated completed subroutine boundary.

2 The finite native word and integer composition

Let $\eta^2+\eta+1=0$ in $\mathbb{F}_4$. For $M\in\mathbb{F}_3^{e\times e}$, define an operator on $\mathbb{F}_4$-valued functions on $\mathbb{F}_3^e\times\mathbb{F}_3^e$ by

$$K_M=\mathcal{F}^{-1}\,\mathrm{diag}_{\xi,\zeta}(\eta^{\xi^T M\zeta})\mathcal{F}.$$

Here $\mathcal{F}f(\xi)=\sum_x \eta^{\xi^T x}f(x)$, and its inverse uses negative exponents. In dimension $m=2e$, its normalizing scalar is $3^{-m}=1$ in characteristic two. Fourier notation proves identities; these transforms are not invoked as fast subroutines. We have $K_MK_L=K_{M+L}$. With $U_Af(x)=f(A^{-1}x)$, conjugation by $A=\mathrm{diag}(G,G^{-T})$ sends K_M to $K_{GMG^{-1}}$ and fixes K_I. The separate reflection $\mathrm{diag}(I-2P,I)$ inverts K_P and fixes K_Q if $PQ=0$. A rank-e paired job uses $2e$ active trits, not e.

Lemma 2 (Finite all-input unit)**.** *There is an effective finite native program, valid at every column multiplicity f. For a fixed positive integer C, its productive and paid profiles are respectively CT and CP, where*

$$T=\{8:18910,22:240,36:240\}$$

and

$$P=\{1:27445,2:10400,3:2200,4:4340,5:7760,6:8120,7:765\}.$$

Every public role has arbitrary independent input. Owned work is initialized, queried and disposed without productive credit. At $s_0=1-1/1300$, $\sum_r P_r r^{s_0}<\sum_k T_k k^{s_0}$.

Proof. Use the 120 integer root lines $2e_i\pm 2e_j$ and $(1,\pm1,\dots,\pm1)$ with even sign parity. Each has norm eight; distinct inner products are $0,\pm4$. Seven coordinate-match bases and eight simplex-code cosets partition them into fifteen orthogonal bases. The explicit two-tree rectangle list in Appendix A partitions all 7560 directed orthogonal contacts once. Its source/target size histogram is

$$\begin{array}{ccc}(1,1):1624, & (1,2):800, & (1,4):232,\\ (2,1):752, & (2,2):108, & (2,4):20,\\ (4,1):208, & (4,2):16, & (4,4):22.\end{array}$$

The binary fitting B has an explicit nine-center factorization. At root scale $\alpha = v/2$, choose the integral E8 simple-root Gram S. Its three arms have lengths $1, 2, 4$, so

$$\det S = 30(2 - 1/2 - 2/3 - 4/5) = 1.$$

For $d_s = (\alpha_s \cdot \beta_i)_i$ the entries are integral, and $B_{st} = 1 + d_s^T S^{-1} d_t$ modulo two. Thus $(1, d_s^T)$ and $(1, S^{-1}d_t)^T$ give $RG = B$. The rectangle incidences give $JV = B + I$.

On each rectangle, grow source flags from leaves to root and target flags from root to leaves. All XOR contacts are then in a common frame. A side of dimensions (a, b) has paid jumps $a, 8 - a - b, b$ and its own rank-eight endpoint on arbitrary data. Each main flag path has jumps $1, 2, 4$. Nine owned centers are gathered at the full frame, queried on disjoint rank-seven and rank-one pieces, scattered and erased. The resulting raw main call is $D\,\mathrm{Lower}(Q^{-1})$ with $Q = K_P$, $D = K_{I-P}$; arbitrary incoming side dirt cancels exactly.

For $a^3 = 1$ the five integral transvections $I + \lambda_i u_i \phi_i$ in Appendix A satisfy $T_5 \cdots T_1 = \mathrm{diag}(a, a, a^2, a^2)$ in $\mathbb{Z}[a]/(a^3 - 1)$. At stages four and five their supports lie in banks zero and two. Apply the complements only on these two banks at those stages. The full endpoints are then rank 36 on two banks and rank 22 on two banks. Helpers remain independent rank-eight outputs. This removes exactly 480 inactive rank-seven calls, not their productive rank mass alone. Including the 45 split owned queries gives the displayed invoice per cover row, with normalized target/paid rank masses 165200 and 165080. The finite cover multiplies both profiles by the same C.

The native identities use commuting projectors and the literal chronology in common frames, at every column multiplicity f. They do not distribute tensor powers through sums. Appendix A specifies the five directions, frames and full invoice. Directed integer bounds for logarithms and exponentials give, after division by 36^{s_0},

$$\text{paid} \le \frac{46549475589303981235}{10133099161583616} < \frac{46549523863948677520}{10133099161583616} \le \text{target}.$$

The certificate evaluates exact finite inequalities, not growing-size timings. Its polynomial-ring multiplication also records all intermediate bank matrices, and its frozen-list verifier exports every rectangle. □

Lemma 3 (Finite uniform column body)**.** *For any fixed $s_0 < z_0 < z < 1$ and $\varepsilon > 0$ there are effective fixed integers N, B, P and an all-input column template for K_{I_N} on B independent arrays using P rank-one queries, including owned queries, with $P/B < \varepsilon N^z$. The same mixed construction has a rational order-three address interpretation.*

Proof. Let $\widetilde{T}_k = CT_k$ and $\widetilde{P}_r = CP_r$ be the actual integer unit multiplicities. For a target composition a put $c(a) = \prod k^{a_k}$ and $u(a) = \binom{|a|}{a} \prod \widetilde{T}_k^{a_k}$. Removing the last colored letter gives $u(b) = \sum_k \widetilde{T}_k u(b - e_k)$. At top rank n choose G maximal with $36\,8^G \le n$. An ordered paid path has multiplier ρ; the target of type k at child $b - e_k$ matches the pending rank-r request since $k\rho rc(b - e_k) = \rho rc(b)$. Its ambient fits because $r < k$. Unmatched sources satisfy $n/36 < s \le n$; only terminal paid exits remain, at rank at most $E(n) = 7(n/36)^{\log_8 7} = o(n)$.

Finish an entire matched child class before resuming its paid parent round. Ordered paths and live owned banks have separate tags. Integer replication clears every smaller batch denominator. Complete each source's disjoint $n - s$ complement before using its rank-s endpoint. With $L(n) = P_n/B_n$ the literal construction gives

$$L(n) \le \sum_s w_s L(n - s) + \sum_e x_e L(e), \qquad \sum_s w_s = 1, \quad \sum_e x_e e^{s_0} \le \sum_s w_s s^{s_0}.$$

Set $\delta = z_0 - s_0 > 0$. Then

$$\sum_e x_e e^{z_0} \le (36E(n)/n)^\delta \sum_s w_s s^{z_0}, \qquad n^{z_0} - \sum_s w_s (n - s)^{z_0} \ge z_0 36^{z_0 - 1} \sum_s w_s s^{z_0}.$$

Strong induction gives $L(n) \le An^{z_0}$. Taking N so that $AN^{z_0-z} < \varepsilon$ supplies any required fixed margin. Rational bounds give an effective cutoff. Splitting, whole-job bases and the all-input unit commute with replacing each coordinate slot by f columns, so the body is valid at every f. The rational interpretation uses the actual root projectors $vv^T/8$ and $I + (R - I) \otimes P$, $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. Its native identities use the same annihilating projectors and scalar XOR word; rational matchers are frozen before a residue prime is chosen. □

3 Ordinary movement from a linear seed

Lemma 4. *For every fixed rational $s_0 < t < 1$ there is a uniform equal-chunk interchange machine costing $O(V(u+1)^t)$ on $[P_0][2^u][G_0][2^u][B_0]$, with arbitrary positive prefix, gap, suffix and payload. Aggregate rectangles of address width at most L cost $O(VL^g)$, $g = 1/(2-t)$.*

Proof. The normalizers and cost induction are summarized here; the complete row, range and tape constructions are in Appendix C. Freeze Lemma 3, then choose prime $q \ge 5$ avoiding its finitely many denominators and witnesses. A generic rank-one address update has split rank two. With E, F the first two head and last two tail selectors, write

$$\begin{aligned} M - I &= \begin{pmatrix} E \\ B_1 \end{pmatrix} (A_1, DF^T), \\ H &= A_1 E + DF^T B_1, \\ K &= (-B_1 + FD^{-1}H)E^T + FD^{-1}A_1(I - EE^T). \end{aligned}$$

Then $KE = -B_1 + FD^{-1}H$, $DF^T K = A_1$, and

$$L(K)ML(-K) = I + \begin{pmatrix} E \\ FD^{-1}H \end{pmatrix} (0, DF^T).$$

Only four coarse fields remain. No H^{-1} is used. Gaussian lowering over the local ring uses unit pivots and at most 43 smaller swaps; use 64. Bad charts use $32N^2$; their fraction is at most $16N^2/q$ by the cleared polynomial bounds and equal high-lift counts.

For a completed output $M = \left(\begin{smallmatrix} X & B_1 \\ C & D_1 \end{smallmatrix}\right)$ with $M^2 + M + I = 0$ and invertible B_1, lower adapters give $[0, B_1; -B_1^{-1}, -I]$, then $J_A = [0, A; A^{-1}, 0]$, $A = -B_1$. The identities $J_A L(-A^{-2}) J_A = U(-I)$ and $J_I = \mathrm{NegHead}\, L(I)U(-I)L(I)$ remove the full matrix weight with two bodies. On bad output rows the bodies still run; $J_I M$ corrects M^2. Choose N, then q, so the normalized child z-moment is $< 3/4$. The normalized stopped recurrence is

$$T(n,w) \le \sum_j c_j T(\lfloor n/N \rfloor, w) + C_N n + C_N (w+1)^b, \qquad T(n,w) = O_N((w+1)^b)\ (n < N),$$

where $b \le 1$ is the predecessor ordinary movement exponent. Both strict moments give $O_N(n + n^z(w+1)^b)$.

Fixed right matchers act affinely on high matrices over each fixed low residue type; predecessor swaps and fixed-unit residue lanes price them. Variable unit pivots use the charged smaller children. The complete q-adic row stock is reserved once, with $O_N(w \log(s+2)) = o(s)$ high digits when $w = \lceil s^{1-z} \rceil$. No new child reapplies the ordinary wrapper or root padding. Crop the private row pad after a full body, then finish predecessor high/remainder swaps before cropping the original binary box.

The ordinary exponent update is $b_{j+1} = z + (1-z)(b_j - z)$, starting from $b_0 = 1$ by explicit four-stream bit-pair exchange. Each fixed successor calls its predecessor, never itself at atom width w. Thus $b_j = z + (1-z)^{j+1} \to z$. The number of tape families and bootstrap passes is fixed before input varies. Finally Euclidean equal-block exchanges above threshold a cost $O(VLa^{t-1})$ and the last short field costs $O(Va)$; take $a = \lceil L^{1/(2-t)} \rceil$. Each whole native aggregate range is padded once and cropped after its completed rectangle. □

4 Full-mask endpoint execution

Lemma 5. *Let $M = \prod_i n_i$, $L = \lceil \log_2(2M) \rceil$, and let the native radices n_i be even, with $L^3 \le n_i \le R_0$, $\log R_0 = \operatorname{polylog} L$, and $V = DM$, $D \ge 1$. For fixed rational $s_0 < t < \ell < 1$ the canonical adjacent-pair $H = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ tensor costs $O(VL^{\max(\ell,g)} \operatorname{polylog} L)$. Its inverse and literal ordinary bit transpose have the same bound.*

Proof. The weighted eight-update tile word of Appendix D flips chosen endpoint bits while restoring its dirty auxiliary. Dock only the selected native factor of u digits, padded to its next binary range, through its actual gap. A high prefix pads whole opposite-factor rows; a low prefix pads each low-factor block. Both cost $O(V)$ and expose $O(u \log R_0)$ bits, not the whole bank. Private pads survive the eight updates and exact repair and are cropped before a phase child.

The quotient-margin bad set $\mathcal{B}$ has fraction $O(u/L^3)$. The ideal T preserves quotient labels, so $T(\mathcal{B}) = \mathcal{B}$. The packed P fixes the private invalid tails and agrees with T on $\mathcal{B}^c$ in the same padded domain. Both are bijections, hence

$$P(\mathcal{B}) = \Omega \setminus P(\mathcal{B}^c) = \Omega \setminus T(\mathcal{B}^c) = T(\mathcal{B}) = \mathcal{B}.$$

This derives the repair-slot invariant rather than assuming it. At a bad output y, sort the whole record by key $T(P^{-1}y)$. Stage a complete item before selecting its stable radix bucket. With K bad items, $b = O(L)$ key bits and opaque width R_{op}, this costs $O(Kb(R_{\mathrm{op}} + b) + V + M_{\mathrm{rec}}L^C)$, including all head returns. Refill only marked holes. Five existing endpoint banks and the native suffix provide dirty selectors, a dock, $O(\log h)$ role bits and $R_{\mathrm{op}} = 2^{\Omega(L)}D$, which pays all fixed metadata polynomials.

Pair endpoint bits into four-state digits; idle state 3 is kept in place. Dirty-selector echoes implement disjoint active-rank SUM, negation and swaps. All masks remain in the array. For a paired rank-e call select the first $2e$ active trits, using prefix $\min(h, \lceil C(2e + \log(L+2)) \rceil)$. Correct only completed logical routes with restored dirty selectors. The exceptional eligible prefix fraction is exponentially small in the full physical universe, without conditioning its distribution on eligibility. The ideal route preserves the mask defining this exception; the truncated route is identity there and equals the ideal elsewhere. The same destination-key repair therefore applies before any linear combination forms a phase. Every child inherits its parent's mask-only enable predicate, conjoined with its local eligibility test. All its charts and translations use that predicate; a disabled parent has identity endpoints on every role. For the fixed body of Lemma 3, each query has actual volume V/B, including owned queries. Reserve rows divisible by $B^{\lceil \log_N(h+2) \rceil}$ once. Children retain every physical mask and opaque lane. Parking/restoring siblings costs $O_N(V)$ at their parent; descendants never scan parked ancestors. Thus

$$F(e; L) \le (P/B)F(\lfloor e/N \rfloor; L) + \operatorname{polylog} L((e + \log(L+2))^t + \operatorname{polylog} L).$$

Unrolling uses $(P/B)/N^t < 1$ after choosing the body exponent below t; it gives $O((e+1)^t \operatorname{polylog} L)$.

For $C = \eta I_3 + \eta^2 J_3$ the Fourier multiplier is η^{ξ^2}. The explicit matrix

$$J = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 2 & 0 & 2 \\ 0 & 2 & 1 & 2 \end{pmatrix} \quad \text{satisfies} \quad \sum_i (Jx)_i^2 = x_0x_2 + x_1x_3.$$

Since $\mathcal{F}U_A f(\xi) = \mathcal{F}f(A^T\xi)$, $U_{J^T}C^{\otimes 4}U_{J^T}^{-1} = K_{I_2}$. Chronologically apply U_{J^T}, then K_{I_2}, then its inverse chart to obtain $C^{\otimes 4}$. For each mask with a active trits, expand $\lfloor a/4 \rfloor$ in base N. Move each consecutive $4N^j$-trit group by two active-coordinate reflections, execute one paired rank-$2N^j$ job with that chart, and undo. Process the at most three leftover active trits directly. All active counts, including zero, are covered; no sector is gathered.

The exact positive $\mathbb{F}_{16}$ word in Appendix E has 384 forward ($C \oplus [1]$) letters and 392 monomials and equals $H \otimes H$. Tensor its letters over a block, not separately per axis. Monomial scalars are generated at source addresses; all four F16 bit planes and pads survive the full word. Completed blocks are binary, fix every borrowed index, and cost the displayed bound including aggregate layouts. Polylogarithmically many blocks cover the axes. The literal completed operator is symmetric; the same algorithm is its bit transpose. Its inverse is two forward applications since $H^3 = I$. □

5 Ordinary transfer and explicit exponent

The explicit outer algorithms of [3] have the following parameter ledger. With $\theta = \max(\ell, g)$, generic/ragged local permutations, controlled tensors, the balanced CRT and local Fourier words cost $O(VL^\theta \operatorname{polylog} L)$. To verify the substitution, adjacent matching slots use two HJ blocks and unit diagonals; ragged fragment topology is independent of switch settings. Controlled scalars are read only at completed subset-kernel boundaries, with the current shuffle gauge inverted. Their complete per-record generator cost is $G(R_0, L)$, so the scalar bill is MG, not MDG. The local formal triadic convolution DAG and dirty Toeplitz replay have only polylogarithmically many such slots. Growing coefficient arithmetic uses the independent short-product backend. No algebraic step uses the numerical value of the older published exponent. The callsite table and sequential generator bounds are proved in Appendix G.

For completeness the ordinary boundary is explicit. The cited host constructs $2n < N_* < 4n$ and its root data in $N_*^{o(1)}$ time. Enlarge its divisor to α' until $D' = \varphi(\alpha') \geq (R_0 + L_0 + 5)^{64}$, where $L_0 = \lceil \log_2 N_* \rceil$, $R_0 = 2^{\lceil \log_2 L_0 \rceil^2}$. Then $m' = N_*/\alpha'$, $m'D' = \Theta(N_*)$ and $1 - D'/\alpha' = O((\log L_0)^2/L_0^3)$. Root transport is the explicit unital old-variable map $Y \mapsto U^J$ followed by the appropriate Jth power; new input words use a variable rename, not that substitution a second time.

For the native grid, $M = m' \prod_{p|m'}(1 + 1/p)$ and $\prod_{p|m'}(1 + 1/p) = 1 + O(L_0^{-2})$. Since $\alpha' > L_0^{64}$ and $\log \alpha' = o(L_0)$, its parameter satisfies $L = \lceil \log_2(2M) \rceil \leq L_0$ and $L = \Theta(L_0)$ for all sufficiently large inputs. Thus $n_i = p_i + 1 > L_0^3 \geq L^3$ and the endpoint hypotheses hold. Ragged CRT and local Fourier slots keep these same full axes.

Map $X \mapsto Y^c Z$, $c = (m')^{-1} \bmod \alpha'$. An ordinary coefficient index $k = r + m'j$ goes to $(r, cr + j \bmod \alpha')$. Split rows by $f = \Phi_{\alpha'}$ and $\psi = (Y^{\alpha'} - 1)/f$ and retain both streams. For $uf + v\psi = 1$ reconstruct as $(A_f v\psi + A_\psi uf) \bmod (Y^{\alpha'} - 1)$. The large stream uses native axes $p_i + 1$, the full dummy grid, and three complete DFTs with CRT-idempotent local root exponents. The complement, with $d = \deg \psi$, packs $Y^j Z^i \mapsto T^{2di+j}$, reduces rows and folds $Z^{m'} = 1$. Its classical cost is $O(N_*(\log L_0)^3/L_0) = o(N_*)$. Setup, scalar arithmetic and top maps cost $N_* \operatorname{polylog} L_0$ or $N_*^{o(1)}$. The complete bill is

$$O(N_* L_0^\theta (\log L_0)^A) + N_* \operatorname{polylog} L_0 + N_*^{o(1)}.$$

Every private crop occurs at its proved completed boundary. Inverse maps, all cyclic coefficients, stream erasure and head returns are included. Since $2n - 2 < N_*$, the first $2n - 1$ returned cyclic coefficients are exactly the ordinary product.

To prove Theorem 1, put $u_0 = 1300/1301$, $z = (s_0 + u_0)/2$ and $d = 1 - z$. Two fixed movement bootstraps give $t = z + d^3$. Since $u_0 - z = 1/3382600$ and $d^3 < 1300^{-3} < 1/3382600$, we have $t < u_0$. Therefore

$$g = 1/(2 - t) < 1/(2 - u_0) = 1301/1302.$$

Choose $t < \ell < g$. Then $\theta = g$ and the positive fixed gap to $1301/1302$ absorbs every fixed secondary logarithm. All choices are made before input length varies. The finite initial range uses the same fixed direct branch, completing the proof.

The ordinary fixed-operand transpose is also explicit. If b is fixed, $(T_b^T c)_i = \sum_j b_j c_{i+j}$ is coefficient $n-1+i$ of $(\sum_j b_j X^{n-1-j})(\sum_{k<2n-1} c_k X^k)$. Run the ordinary algorithm at length $2n-1$, emit its full product and crop indices $n-1$ through $2n-2$. Reversal, padding, cropping and returns add $O(n)$.

Computational certificates. The accompanying exact certificates verify the finite field identities, orthogonal rectangle partition and strict rank inequality used in Lemma 2. The asymptotic conclusion follows from the recursive constructions and charged interfaces, not from finite timing tests. Their scope is distinct from independent mathematical review or a complete formal proof of the main theorem.

A The finite all-input word

A.1 Root lines, projectors and binary incidence

Take one representative of each of the 120 lines $2e_i \pm 2e_j$ $(i < j)$ and $(1, \pm 1, \ldots, \pm 1)$ with even sign parity. Each vector v_s has squared norm eight. Distinct representatives have inner products $0, \pm 4$. Put $P_s = v_s v_s^T/8$ over $\mathbb{Q}$, or reduce the same expression over $\mathbb{F}_3$. Thus P_s is a rank-one idempotent; orthogonal root lines give mutually annihilating idempotents in both interpretations.

Let $A_{st} = 1$ for distinct orthogonal lines and zero otherwise. Over $\mathbb{F}_2$ put $B = A + I$. At scale $\alpha_s = v_s/2$ choose

$$\begin{aligned} &\beta_1 = (1,-1,-1,-1,-1,-1,-1,1)/2, \qquad \beta_2 = e_1 + e_2, \\ &\beta_3 = -e_1 + e_2, \quad \beta_4 = -e_2 + e_3, \quad \beta_5 = -e_3 + e_4, \\ &\beta_6 = -e_4 + e_5, \quad \beta_7 = -e_5 + e_6, \quad \beta_8 = -e_6 + e_7. \end{aligned}$$

Their Gram matrix S is the E8 tree matrix with diagonal two and three arms of lengths $1, 2, 4$. A length-l arm has determinant $l + 1$ and inverse endpoint entry $l/(l + 1)$. Taking its Schur complement gives

$$\det S = 30(2 - 1/2 - 2/3 - 4/5) = 1.$$

The vectors $d_s = (\alpha_s \cdot \beta_i)_i$ are integral: two half roots have even Hamming distance, and integer/half-root products are integral. Since the β_i form a basis, $\alpha_s \cdot \alpha_t = d_s^T S^{-1} d_t$. The integral matrix S^{-1} therefore gives the explicit binary factors

$$R_s = (1, d_s^T), \qquad G_t = (1, S^{-1} d_t)^T, \qquad RG = B.$$

The equality follows entrywise: the root inner product is even on the diagonal and on orthogonal pairs, and odd on every other pair.

A.2 Fifteen bases and the rectangle list

Index the eight coordinates by $x \in \mathbb{F}_2^3$. For each nonzero a, the four pairs $\{x, x + a\}$ give the eight orthogonal roots $2e_x \pm 2e_{x+a}$. These seven bases partition the 56 sparse roots. The binary simplex code $(a \cdot x)_{x \neq 0}$ has eight words; its nonzero words have weight four. Its eight cosets in the even-weight length-seven code therefore give eight orthogonal bases of sign vectors with coordinate zero fixed positive. Take the least unused integer word as each coset representative. This partitions the remaining 64 root lines.

Give each basis a balanced tree on eight positions. Enumerate the 21 flags by choosing $a, b \in \{1, \ldots, 7\}$ with $b \neq a$ and $b < (b \operatorname{xor} a)$, and then the least c outside their span. Relabel position x by $4(a \cdot x) + 2(b \cdot x) + (c \cdot x)$. In the base order above use the following zero-based source and target flag indices:

$$\begin{aligned} f_X &= (9, 0, 0, 4, 0, 16, 1, 9, 15, 9, 9, 1, 13, 1, 15), \\ f_Y &= (10, 2, 0, 10, 4, 15, 0, 10, 16, 11, 14, 0, 14, 10, 15). \end{aligned}$$

The accompanying file `checks/expected/frozen_rectangles.json` lists the 3782 rectangles. Each source and target set is a node of the corresponding tree. The verification algorithm checks that each rectangle consists entirely of orthogonal pairs and that its directed pairs are disjoint from all earlier rectangles; finally it checks that every directed orthogonal pair has been covered. All sets have at most eight elements, so this is a finite exact certificate with no asymptotic premise. Its histogram is the one in Lemma 2; $\sum abS_{a,b} = 7560$.

Let $V_{e,t} = 1$ when t is in rectangle e's source set and $J_{s,e} = 1$ when s is in its target set. Then $JV = A$ and

$$RG + JV = I \quad \text{over } \mathbb{F}_2. \tag{1}$$

This binary identity concerns payload additions. It is not reduced from a rational identity.

A.3 The common-frame program

Write $\mathcal{K}_U$ for the paired phase associated with a projector P_U. A field in frame F holds F times its logical value. Initially assign data field X_t frame $\mathcal{K}_{P_t}$ and fields Y_s, Z_e frame I. This assignment is notation: its logical X_t value is $\mathcal{K}_{P_t}^{-1} X_t$, and it requires no unpaid initial query. The following chronological program uses independent arbitrary Y and side inputs Z.

1. Subtract JZ from Y, all in frame I.

2. Visit source-tree nodes from leaves to root. Grow participating X_t fields to the node frame $\mathcal{K}_U$, advance every contacting side to that same frame, and add its V-row sum of X_t. Each contact therefore takes place in a common frame. At the root all X fields are in frame $\mathcal{K}_I$.

3. Form the nine G-row increments as owned fields. On each apply $\mathcal{K}_I^{-1}$, implemented on disjoint rank-seven and rank-one pieces, scatter it by R to the still unadvanced Y, and erase it. All inverse queries are paid.

4. Visit target-tree nodes from root to leaves. At a node of span W, advance its Y_s fields and contacting sides to $\mathcal{K}_{I-P_W}$, then add the side through J. At a leaf the Y_s frame is $D_s = \mathcal{K}_{I-P_s}$.

5. Advance every side to $\mathcal{K}_I$ and subtract its V-row sum of the full-frame X. Its endpoint is $\mathcal{K}_I Z_e$.

To verify the word, denote the incoming logical X values by x. After the source sweep side e holds $\mathcal{K}_U(Z_e + (Vx)_e)$. Its contribution to a target s after the target sweep is $D_s(Z_e + (Vx)_e)$. The first subtraction cancels its arbitrary incoming value. The remaining logical target increment is $(RG + JV)x = x$ by (1). Thus the raw main endpoint is

$$\operatorname{diag}(D, D) \begin{pmatrix} I & 0 \\ Q^{-1} & I \end{pmatrix}, \qquad Q_s = \mathcal{K}_{P_s}, \quad D_s = \mathcal{K}_{I-P_s}.$$

Every side has its own full rank-eight endpoint. A side of dimensions (a, b) pays jumps $a, 8 - a - b, b$; rank zero is the identity. Each main tree path pays $1, 2, 4$. This verifies arbitrary side inputs, not just zero-initialized side fields.

The same chronology applies to the rational address permutations $I + (R_3 - I) \otimes P_U$, where $R_3 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. Payload additions are still binary, so (1) is unchanged. In inverse stages all frames, including complementary and helper frames, are inverted. Replacing every coordinate slot by f columns preserves the projector relations and each common-frame contact; it does not distribute a tensor power over a sum.

A.4 Five stages and the partial endpoints

For $a^3 = 1$, set $T_i = I + \lambda_i u_i \phi_i$ in the ring $\mathbb{Z}[a]/(a^3 - 1)$, with chronological order

i	u_i^T	ϕ_i	λ_i
1	$(0, 0, 1, 1)$	$(1, 1, 0, 0)$	$a - 1$
2	$(0, 1, -1, -1)$	$(-1, -1, 0, -1)$	$a^2 - 1$
3	$(0, 1, -1, 0)$	$(-1, 0, 0, -1)$	$a - 1$
4	$(1, 0, -1, 0)$	$(-1, 0, -1, 0)$	$a^2 - 1$
5	$(1, 0, 0, 0)$	$(0, 0, -1, 0)$	$a - 1.$

Each $\phi_i u_i = 0$, and coefficientwise multiplication gives

$$T_5 T_4 T_3 T_2 T_1 = \operatorname{diag}(a, a, a^2, a^2). \tag{2}$$

The three coefficients of every entry at each intermediate product are supplied by `checks/expected/finite_certificate.json`. They give an exact integer polynomial identity, rather than a test on sampled eigenvalues.

Set $a = Q^{-1}$. A fixed bank basis turns the raw shear into each listed direction; the free constant part of the shear is subtracted by a binary bank update. Use five windows sharing the rank-one projector P and having disjoint rank-seven complements. Initially apply each complement D_i on all four banks. For stages four and five instead use the unimodular bases

$$S_4 = [-e_0, e_0 - e_2, e_1, e_3], \qquad S_5 = [-e_2, e_0, e_1, e_3]$$

and apply D_i on their two active banks only. Both transvections preserve $E = \mathrm{diag}(1, 0, 1, 0)$, so the endpoint is

$$D_1 D_2 D_3 (I - E + E D_4 D_5)\,\mathrm{diag}(a, a, a^2, a^2).$$

The partial projector reflection on banks zero and one changes their a to $a^{-1} = Q$ and fixes every complement. Banks two and three already have $a^2 = Q$. Thus banks zero and two receive rank 36 and banks one and three rank 22; the other slots are identity. The bank bases and the five window placements are fixed slot maps. To match equal-rank phase jobs, index copies by the full finite $\mathrm{GL}_{36}(\mathbb{F}_3)$ cover and use right multiplication for these placements. This permutes copies and multiplies productive and paid counts by the common fixed integer $C = |\mathrm{GL}_{36}(\mathbb{F}_3)|$. For rational addresses take C schematic copies with fixed right-route symbols; their full residue cover is constructed after precompilation in Appendix C, without a finite rational matcher group.

There are five independent sides per rectangle, hence 18910 rank-eight outputs, and 240 main outputs at each of ranks $22, 36$. Sides contribute five jumps at each of $a, 8-a-b, b$ per rectangle; main flag paths contribute 1200 calls at each of ranks $1, 2, 4$. The partial-bank schedule leaves 720 inactive rank-seven calls. The 45 owned increments contribute one call each at ranks one and seven. These give the normalized profiles T, P of Lemma 2. The actual covered program has the integer profiles CT, CP. Multiplication by C preserves the strict moment and rank thresholds; it is included in the compiler's integer colors and finite constants.

A.5 The strict moment certificate

For a rational $x \in [1, 2]$, put $y = (x-1)/(x+1) \le 1/3$. Then

$$\log x = 2 \sum_{j=0}^{47} \frac{y^{2j+1}}{2j+1} + E, \qquad 0 \le E \le \frac{18 y^{97}}{8 \cdot 97}.$$

Normalize each $36/r$ by powers of two and use these directed bounds with binary scale 2^{48}. For $h = \log(36/r)/1300$ use

$$1 + h + h^2/2 \le e^h \le 1 + h + \frac{h^2}{2(1 - h/3)}.$$

Since $(r/36)^{s_0} = (r/36)e^h$, outward integer rounding gives the two rational endpoints displayed in Lemma 2. Their numerator difference is $48274644696285 > 0$. The fixed checker prints all per-rank intervals, so the strict moment is independently reproducible with exact arithmetic. No growing-size running time is inferred from this finite calculation.

B Integer composition of different-rank jobs

We prove the column-template claim used by Lemma 3. The argument is a specialization of the mixed-job and finite precompilation mechanisms of [6, 7]. All jobs here have the exact all-input word of Appendix A.

B.1 Interfaces and whole-job matching

A rank-r paired job is K_{I_r} on $2r$ selected trits and identity on every spectator. Its f-column version is $K_{I_r \otimes I_f}$. Disjoint completed jobs split by $K_{I_{a+b}} = K_{I_a \oplus 0} K_{0 \oplus I_b}$. An equal-rank idempotent is matched by bases of its image and kernel: the coordinate permutation $\mathrm{diag}(G, G^{-T})$ conjugates it to the required canonical job and preserves K_I. Every such normalizer is a fixed finite slot map lifted by I_f.

For the rational address interpretation, replace the phase by $I + (R_3 - I) \otimes P$. Equal-rank completed jobs are similar by rational bases. Inverse R_3 modules are rationally similar to R_3, and spectators are matched as whole fixed spaces. There is no assumption that the rational matcher group is finite. The finite constructor records its finitely many right-route symbols for the later residue cover.

B.2 A finite integer flow

Let T_k and P_r in this appendix denote the actual integer profiles $CT_k^{\mathrm{norm}}, CP_r^{\mathrm{norm}}$ from Lemma 2, so $r \le 7 < 8 \le k \le 36$. For a target-type multi-index a put

$$c(a) = \prod_k k^{a_k}, \qquad u(a) = \binom{|a|}{a} \prod_k T_k^{a_k}.$$

Regard its $u(a)$ copies as colored words of composition a; the letter of type k has T_k possible colors. Removing the last letter gives the bijection

$$u(b) = \sum_{k : b_k > 0} T_k u(b - e_k). \tag{3}$$

For top rank $n \ge 36$, take G maximal with $36\,8^G \le n$. At depth $d \le G$, retain each ordered paid-round path and its multiplier ρ, the product of its preceding paid ranks. The class of composition a, $|a| = G - d$, contains $u(a)$ units at column multiplicity $\rho c(a)$ whenever $36\rho c(a) \le n$. For a pending rank-r request at parent composition b, the child of type k has column multiplier $\rho r c(b - e_k)$ and productive rank

$$k \rho r c(b - e_k) = \rho r c(b).$$

Its ambient fits because $r/k < 1$. Equation (3) matches every request to a distinct whole child output, including its color. Missing parent slots become new independent source arrays.

Every such source has rank $n/36 < s \le n$. Indeed a missing parent fit gives $36\rho_{\mathrm{parent}} c(b) > n$ and hence $s > rn/36 \ge n/36$; the child's fit supplies the upper bound. At depth zero maximality of G gives the same lower bound. Only depth-G requests remain unmatched. Their rank is at most

$$E(n) = 7^{G+1} \le 7(n/36)^{\log_8 7} = o(n). \tag{4}$$

An ordered path is not merged with another path merely because their multipliers coincide.

The schedule is chronological and acyclic. Suspend a class at its next paid round, execute the entire matching child class, restore its routes and resume the parent. A pending owned increment is a child input of its actual current value; it is not counted again as an independent source. Each suspended owner has a distinct tag, and its erase occurs only after return.

B.3 Clearing denominators and paying complements

Inductively suppose a rank-$e < n$ template uses B_e independent inputs and P_e rank-one queries. A direct template has $B_e = 1$, $P_e = e$; set $B_0 = 1, P_0 = 0$. Let S_s be the source counts in the finite flow and t_e its terminal exit counts. Replicate the entire flow by a common positive multiple of every B_{n-s} and B_e that it uses. All smaller-template batch counts are now integers.

Complete each source's disjoint rank-$(n-s)$ complement before its rank-s flow endpoint, and complete each exit by the smaller rank-e template. These are actual paid source and exit jobs. Every original source then receives the full rank-n endpoint.

Writing $S = \sum_s S_s$, $w_s = S_s/S$, $x_e = t_e/S$ and $L(n) = P_n/B_n$, this construction gives

$$L(n) \le \sum_s w_s L(n-s) + \sum_e x_e L(e), \qquad \sum_s w_s = 1. \tag{5}$$

The strict unit moment, multiplied by each unit's column multiplier to the power s_0, is preserved under repetition. Matched internal requests cancel against their productive endpoints, leaving

$$\sum_e x_e e^{s_0} \le \sum_s w_s s^{s_0}. \tag{6}$$

This is an inequality on the finite constructed flow, including all owned requests; it does not promote a smaller helper to a full output.

Choose $s_0 < z_0 < z < 1$ and put $\delta = z_0 - s_0$. By (4) and $s > n/36$,

$$\sum_e x_e e^{z_0} \le (36E(n)/n)^{\delta} \sum_s w_s s^{z_0}.$$

For $v = s/n \ge 1/36$, concavity gives $1-(1-v)^{z_0} \ge z_0 v \ge z_0 36^{z_0-1} v^{z_0}$. Consequently

$$n^{z_0} - \sum_s w_s (n-s)^{z_0} \ge z_0 36^{z_0-1} \sum_s w_s s^{z_0}.$$

The first coefficient tends effectively to zero. Choose an effective cutoff where it is smaller than the second coefficient, and enlarge A so that $L(n) \le An^{z_0}$ on the finite direct range. Strong induction in (5) proves the same bound for every n. Each complement and exit rank is strictly smaller than n. For any prescribed fixed positive rational ε, take N with $AN^{z_0-z} < \varepsilon$. Then $P_N/B_N < \varepsilon N^z$.

The common cover multiplicity changes the integer color counts and the effective constants A, N, B_N, P_N. It leaves the strict unit moment, source-rank thresholds and exit exponent unchanged. All replication factors, slot maps and word lengths are fixed finite integers once N is fixed. Induction through the constructor applies at every column multiplicity f, because the unit, splitting and whole-job matching were proved at every f. Its setup can therefore be encoded in one fixed machine, or generated at a fixed charged cost absorbed by the finite direct cutoff. No advice depending on the eventual ordinary input is used.

C Full-cover movement and its ordinary boundary

The finite-cover and affine-routing method develops the constructions of [4, 7]; the full-weight normalization and ordinary boundary below are proved for the present rational body.

This appendix proves the movement interface of Lemma 4 from the finite template of Lemma 3. An *atom* is an address field of w base-q digits, of range q^w. A coarse field consists of several such atoms; the payload and every other coordinate remain spectators during an ordinary interchange. If a call has active input array length V, all costs below are divided by this actual V. In particular V includes retained pads and opaque payloads in the current call. Cells in a parked ancestor frame are not part of the active call and are not traversed by its descendants; the parent's scans that park, restore, and erase such a frame are charged to the parent's own $O_N(V)$ node toll.

C.1 The rational address body and its full-cover endpoint

Fix the finite body supplied by Lemma 3, with rank $N \geq 2$, B independent productive arrays, and P paid rank-one queries. Put

$$a = P/B.$$

This is a real number: cover rows and role multiplicities may give fractional normalized coefficients. A rank-one query sees the whole current record array divided by B, hence has volume exactly V/B; this includes owned queries. Let z_0, z be fixed rationals with

$$s_0 = 1 - \frac{1}{1300} < z_0 < z < 1.$$

The body lemma allows N to be chosen so that $a < \varepsilon N^z$ for any prescribed fixed $\varepsilon > 0$.

The address action is over a rational vector space; the payload action is over $\mathbb{F}_2$. These are separate rings. Write R for the fixed two-dimensional rational matrix of order three used in the address representation. The frozen native body realizes a full address operator M_g on every cover row g, with

$$M_g^2 + M_g + I = 0, \qquad M_g^3 = I. \tag{7}$$

The equality is an identity of address matrices and is used only for composition of address maps. The corresponding permutation of payload arrays is not asserted to satisfy an additive polynomial identity. All adapters and native jobs are executed on complete role arrays, with arbitrary independent initial contents.

After freezing the finite body and its finite rational matcher list, choose a prime $q \geq 5$ avoiding every denominator and every nonzero witness coefficient occurring below. For $w \geq 1$, let

$$Q = q^w, \qquad \mathcal{G}_w = \mathrm{GL}_{2N}(\mathbb{Z}/q^w\mathbb{Z}), \qquad d = 2N.$$

Each cover row is indexed by $g \in \mathcal{G}_w$ and carries its own $M_g = gTg^{-1}$, where T is the frozen full order-three address matrix. The cover is the full group, not a subgroup presumed to contain the rational matchers. Every fixed matcher acts by a permutation of this cover.

For a paid leaf, the matrix $M - I$ has split rank two over $\mathbb{Q}$. On a good chart, unit-lower row and column operations normalize its selected cross block to EDF^{T}, where E, F select two coordinates and D is diagonal with unit entries. Write

$$M - I = UV, \quad U = \begin{pmatrix} E \\ B_0 \end{pmatrix}, \quad V = (A_0, DF^{\mathsf{T}}), \quad H = VU = A_0E + DF^{\mathsf{T}}B_0,$$

and set

$$K = (-B_0 + FD^{-1}H)E^{\mathsf{T}} + FD^{-1}A_0(I - EE^{\mathsf{T}}).$$

The selector identities $E^{\mathsf{T}}E = F^{\mathsf{T}}F = I_2$ give, by direct multiplication,

$$KE = -B_0 + FD^{-1}H, \qquad DF^{\mathsf{T}}K = A_0,$$

and therefore, with $L(K) = \left(\begin{smallmatrix} I & 0 \\ K & I \end{smallmatrix}\right)$,

$$L(K)ML(-K) = I + \begin{pmatrix} E \\ FD^{-1}H \end{pmatrix}(0, DF^{\mathsf{T}}). \tag{8}$$

The residual in (8) is supported on four coarse fields, regardless of N. Only the selected unit pivots are inverted; H is not inverted. Thus four-field Gaussian lowering applies over the local ring $\mathbb{Z}/q^w\mathbb{Z}$.

We record the chart count. For each leaf, require its selected northeast one-by-one and two-by-two minors to be units. A witness for their simultaneous nonvanishing is obtained by

taking E to be the first two head columns, F the last two tail columns in reverse order, and $H_0 = R - I_2$; with $U = (E^\mathsf{T}, (FH_0)^\mathsf{T})^\mathsf{T}$ and $V = (0, F^\mathsf{T})$, one has $VU = H_0$, $\det H_0 = 3$, and $I + UV$ is rationally similar to $R \oplus I$. The selected minors are nonzero (indeed units after the fixed conjugacy denominators are avoided). For the full output require the entire N-by-N upper-right block to be invertible; the witness $R \otimes I_N$ has upper-right block $-I_N$.

After clearing the fixed conjugacy denominators, each entry of $gT\,\mathrm{adj}(g)$ has degree at most d. The product of the two selected leaf-minor polynomials has degree at most $3d = 6N$; the full-output determinant has degree at most $Nd = 2N^2$. The number of invertible matrices in $\mathrm{GL}_d(\mathbb{F}_q)$ is at least $3/4$ of all matrices, since

$$\prod_{i\geq 1}(1 - q^{-i}) \geq 1 - \sum_{i\geq 1} q^{-i} \geq 3/4.$$

The polynomial zero bound, followed by conditioning on invertibility, therefore gives bad fractions at most $12N/q$ for a leaf and $4N^2/q$ for the full output. We use the common bound

$$\delta_{\mathrm{bad}} = 16N^2/q. \tag{9}$$

Every low matrix $g_0 \in \mathrm{GL}_d(\mathbb{F}_q)$ has exactly $q^{d^2(w-1)}$ lifts, so these same fractions apply at every w. The average is taken separately for each leaf type and for the root output type; no union bound over occurrences in the finite word is used.

On a good leaf, unit-pivot Gaussian elimination on four fields costs at most

$$4 + 2 \cdot 4 \cdot 3 + 5 \cdot 3 = 43 < G := 64$$

ordinary interchanges. Here an upper addition is one lower addition conjugated by two swaps. A paired unit scaling uses five swaps by

$$J_a = \mathrm{NegLast}\; U(a)L(-a^{-1})U(a), \qquad U(a) = J_1 L(a) J_1,$$

followed by J_1; fixed signs and lower additions are streamed. On a bad leaf or bad root row, Gaussian elimination on all $2N$ fields uses at most $F_N := 32N^2$ ordinary interchanges. No nonunit is inverted. These interchanges are the only recursive calls; all other row/column operations in the fixed normalizers are streamed affine updates.

C.2 Removing the full matrix weight

The good-root adapters must remove the variable matrix weight of the full cover, not merely a canonical representative. Write

$$M = \begin{pmatrix} X & B_0 \\ C & D_0 \end{pmatrix}, \qquad M^2 + M + I = 0,$$

and assume B_0 is invertible. The block identities imply $D_0 = -I - B_0^{-1} X B_0$ and direct multiplication yields

$$L(B_0^{-1}X)ML(-B_0^{-1}X) = \begin{pmatrix} 0 & B_0 \\ -B_0^{-1} & -I \end{pmatrix}.$$

After a lower adapter and a head negation this is

$$J_A = \begin{pmatrix} 0 & A \\ A^{-1} & 0 \end{pmatrix}, \qquad A = -B_0.$$

For this full, possibly variable A, block multiplication gives

$$J_A L(-A^{-2}) J_A = U(-I), \qquad J_I = \mathrm{NegHead}\, L(I)U(-I)L(I). \tag{10}$$

Consequently two completed calls to the same body, with the lower adapters in (10), cancel the full weight and leave the prescribed canonical endpoint. The body restores all cover and role fields before returning, so the second call sees the same g and the same $A(g)$. No variable unpaired scaling is delegated to a child.

On bad root rows, set every wrapper adapter to the identity, but still execute both complete bodies on every row. Their endpoint there is M_g^2. After the completed calls, apply the fixed correction $J_I M_g$ by the F_N-interchange fallback; since $M_g^3 = I$, this gives J_I. This branch is taken only at completed endpoints, because intermediate right routes can mix row labels.

The normalized child coefficient is therefore bounded by

$$\mu_z \le \frac{2Ga + \delta_{\mathrm{bad}} F_N (2a+1)}{N^z}. \tag{11}$$

The $2Ga$ term counts the two bodies on every row; the bad-leaf fallback budgets are included in the two a contributions, and the final $\delta_{\mathrm{bad}} F_N$ is the one completed-root correction. The extra factor in $(2a+1)$ is a conservative allowance for the full bad-row branches while preserving the same volume normalization. More precisely, coefficients in (11) are real normalized totals: the fraction of rows in a chart class multiplies its number of child calls, while each query volume is V/B. This is not an integer number of calls per individual row. Owned queries are included in P and therefore in a.

Choose N using Lemma 3 with $\varepsilon = 1/512$, so $a < N^z/512$. Then $2Ga/N^z < 1/4$. Choose q after the finite body and all witnesses are frozen, large enough for (9), prime avoidance, and

$$q > 2048 N^4 (2P + B).$$

Using $N^z \ge 1$, $B \ge 1$, and $a = P/B$, this implies $\delta_{\mathrm{bad}} F_N (2a+1)/N^z < 1/4$. Thus $\mu_z < 1/2$; in particular $\mu_z < 3/4$. Since $N > 1$ and $z < 1$, $\mu_1 \le \mu_z < 1$.

C.3 Fixed rational routes and batched descriptors

For every fixed rational matcher S in the frozen word, all denominators are units modulo q^w. Write a cover index as

$$g = g_0 + qG_1, \qquad g_0 \in \mathrm{GL}_d(\mathbb{F}_q), \quad G_1 \in (\mathbb{Z}/q^{w-1}\mathbb{Z})^{d^2}.$$

The low type g_0 ranges over a fixed finite set. On each such type, right multiplication by S induces a fixed affine bijection

$$G_1 \longmapsto G_1 S + c(g_0) \pmod{q^{w-1}}. \tag{12}$$

The fixed linear part in (12) factors into a fixed number (depending only on N, q, S) of coordinate swaps, shears and fixed-unit scalings. Lower shears are rotations of complete target fibers. Upper shears use the previously compiled ordinary interchange on fields of width $O(w)$. Fixed scalings use a fixed number of residue lanes. These route operations are independent of input data and of the cover index except through its streamed descriptor; variable Gaussian pivots are still charged by the child calls counted in (11). No finite subgroup of $\mathrm{GL}_d(\mathbb{Q})$ is assumed.

Here is the array shape that supplies the target fibers. At recursion depth j, split the unused numeric row stock as

$$[R_j] \times [B] \times [|\mathrm{GL}_d(\mathbb{F}_q)|] \times \prod_{i=1}^{d^2} [q^{w-1}].$$

After this row factor the array contains two complete n-atom address fields, an arbitrary intervening range, spectator coordinates and an opaque bit payload. These are Cartesian factors of its lexicographic index, not headers appended to each payload record. The B factor selects independent source streams; the next factors index g_0 and the d^2 entries of G_1. Thus fixing a

chart, a role, every spectator and all but any one target atom leaves the complete target fiber $[q^w]$. Good and bad chart conditions depend only on g_0, not on that target coordinate. Owned arrays have the same index set even when their initialized values are zero.

Right routing permutes only the chart factors and preserves this product shape at completion. A normalizer updates a target atom by a bijection depending on the other factors, so its completed array again has every target position. At a recursive call the consumed B and chart factors are fixed while the independent next row factor and all smaller target atoms remain complete. Induction proves the target-fiber property at every depth; it is independent of the payload values.

Restriction to a good or bad chart class fixes only the current low type g_0. Packing those rows in their inherited order preserves R_j as an independent complete Cartesian factor. Right routing acts on the current g factor alone. Hence the next $W(w)$ factor extracted from R_j remains uniform conditional on every ancestor chart choice. Current cover labels remain spectators and are not divided from query volume a second time. The scans extracting and restoring a chart class cost $O_N(V)$ at its parent and are included in that parent's node toll.

An odometer enumerates the chart indices. Hold one current fiber header on work tapes and generate its offsets before scanning that fiber; do not copy chart digit strings into each record. The finite g_0 table, fixed-size rational matrices and low-type pivot choices are fixed data. Matrix arithmetic modulo q^w uses schoolbook operations and unit inversion with polynomial bit cost in w. Ancestor headers are stack entries on a fixed tape set. Their depth is $O_N(\log(s+2))$; on the chosen root families $s = O(w^{1/\delta_0})$, so even redundant sequential scans and recomputation cost $C_N(w+1)^r$ for a fixed integer r. This also bounds normalizer coefficients and offset endpoints. They are generated once per complete target fiber, not once per payload bit. There is one batched recursive invocation, not a separately allocated program per high matrix. Since each fiber has q^w positions, for every fixed integer r, there is an effective $C_{q,r}$ with

$$(w+1)^r \le C_{q,r} q^w \qquad (w \ge 1), \tag{13}$$

so fixed-degree descriptor work is absorbed by that fiber's record scan. All counters use fixed-width digit lanes with ripple carries; resets and head returns are charged to the full fiber scan. The stack depth is at most $D = O_N(\log(s+2))$, and its descriptor volume is $O_N(w \log(s+2))$ digits. This is the only growing stack dimension and it is represented on a fixed number of tapes with separators; no tape is allocated per chart, axis, or recursion depth.

For completeness, multiplication by a fixed integer unit also has an explicit streaming implementation. Let $Q = q^w$ and let $a > 0$ be fixed with $(a, q) = 1$. Partition $x \in [0, Q)$ into the at most a intervals on which $j = \lfloor ax/Q \rfloor$ is constant. On the jth interval the image $y = ax - jQ$ is increasing and satisfies $y \equiv -jQ \pmod a$. Since $(Q, a) = 1$, these residues are distinct. A fixed a-lane merge therefore emits the records in increasing y order, implementing $x \mapsto ax \pmod Q$ in $O_a(Q)$ record moves. For the inverse, partition the input y by the residue $r_j = -jQ \pmod a$; in that lane set $x = (y + jQ)/a$. Its image interval is $[jQ/a, (j+1)Q/a)$, so concatenating the a lanes in order emits every $x \in [0, Q)$ once, again in $O_a(Q)$ moves. Numerator, denominator and sign lanes compose to implement any fixed rational unit route. The lane count depends only on the frozen rational coefficient. This construction does not implement a variable unit for free.

C.4 The stopped recurrence and actual row stock

Let $b \le 1$ be the exponent of a previously compiled ordinary interchange supplier. Fix an admissible root width s and atom width $w = \lceil s^{\delta_0} \rceil$, with $0 < \delta_0 < 1$. Let $T(n, w)$ be the total bit time divided by the actual active volume for the covered body on n atoms per field, averaged with the real row weights described after (11). At a node, put $f = \lfloor n/N \rfloor$. The recursive children are calls to this same covered-body construction on strictly smaller n; the predecessor supplier is used only for fixed routes, base cases, and the fewer than N remainder atoms. There are finitely many fixed tape families: input/output, current roles, owned work,

repair/padding, and descriptor/parking stacks. Before a child, the parent parks only its current sibling records and copies the child's active stream into an adjacent frame. The child cannot scan parked ancestors. On return, the parent restores that sibling and erases the frame. With fixed N and a fixed number of child types, all such parent operations cost $C_N n$ per node, not an additional multiplier on the descendant time.

The number of fresh cover/role labels per depth is

$$W(w) = B\,|\mathrm{GL}_{2N}(\mathbb{F}_q)|q^{4N^2(w-1)}.$$

Take $D = \lceil \log_N(s+1) \rceil + 1$ and choose k with $q^k \geq W(w)^D$. Reserve k high q-adic digits in the head row field. Pad that row field once to a multiple of $W(w)^D$; the expansion factor is less than two. Every descent consumes one fresh $W(w)$ factor and restores its labels only at the completed endpoint. The factor is divided from the parent stock, not recreated in a child. Since N, q, B are fixed,

$$k = O_N(w \log(s+2)) = o(s), \qquad n = \left\lfloor \frac{s-k}{w} \right\rfloor.$$

The tail high field and all non-head fields remain spectators during the body. After the complete body fixes the reserved head row, crop artificial row padding. Use the predecessor supplier on the high head/tail pieces (width k) and the low remainder pieces (width less than w); only after these whole-field interchanges crop the original binary box. The initial ordinal padding of each full binary address range to the least $q^s \geq 2^e$ has factor less than q^2, a fixed constant.

The completed-call recurrence is

$$T(n,w) \leq \sum_j c_j T(\lfloor n/N \rfloor, w) + C_N n + C_N (w+1)^b \quad (n \geq N), \tag{14}$$

$$T(n,w) \leq C_N (w+1)^b \quad (1 \leq n < N). \tag{15}$$

The $c_j \geq 0$ are real normalized multiplicities, not integer call counts; their sum at exponent z is bounded by (11). Writing $\mu_1 = \sum_j c_j/N$ and $\mu_z = \sum_j c_j/N^z$, we have $\mu_1, \mu_z < 1$. Strong induction on n proves

$$T(n,w) = O_N(n + n^z (w+1)^b). \tag{16}$$

Indeed, if $T(m,w) \leq Am + B_0 m^z (w+1)^b$ for smaller m, the child sum in (14) is at most $A\mu_1 n + B_0 \mu_z n^z (w+1)^b$. Choose $A \geq C_N/(1-\mu_1)$ and $B_0 \geq C_N/(1-\mu_z)$, enlarging both to cover (15). The same induction applies to the real weighted sum because all coefficients are nonnegative.

Substituting $n = O(e^{1-\delta_0})$ and $w = O(e^{\delta_0})$ into (16), and charging the high/low cleanup with the predecessor, gives

$$O\Big(V\left[e^{1-\delta_0} + e^{(1-\delta_0)z+\delta_0 b} + e^{\delta_0}\log(e+2) + 1\right]\Big). \tag{17}$$

The exponents in (17) are kept separately: no absorption is asserted without comparing them. Set $\delta_0 = 1 - z$. For $b > z$, the middle exponent is

$$(1-\delta_0)z + \delta_0 b = z^2 + (1-z)b = z + (1-z)(b-z) > z,$$

whereas the first exponent is z and the reserve exponent is $1 - z < z$. The logarithm in the reserve term is absorbed in the positive gap between $1 - z$ and z. Thus the exact successor exponent is

$$b' = z + (1-z)(b-z), \qquad b_0 = 1,$$

It follows that $b_j = z + (1-z)^{j+1}$. For any fixed $t > z$, a finite j has $b_j < t$. Compile stages in order. Stage j uses stage $j-1$ only for fixed routes, base/remainder work, and ordinary interchanges inside a covered node. The covered children themselves use the same

stage-j construction at strictly smaller n, as in (14); well-founded induction on n defines that recursion. There is no call to stage j at the atom-width parameter w. Therefore the number of stages, tapes and all constants are fixed before the input width varies, and this is a noncircular finite bootstrap. A zero-width interchange is the identity and has a direct constant-per-input-bit implementation, outside the positive-n recurrence.

D Native endpoint tensors on ordinary tape addresses

Let $\Omega = \prod_{i=1}^{a}[n_i]$, $M = \prod_i n_i$, $L = \lceil \log_2(2M) \rceil$, and $V = MD$, where $D \geq 1$ is the number of payload bits per record. The radices are even and satisfy $L^3 \leq n_i \leq R$, $\log R = \operatorname{polylog} L$. Write $\tau = t$ for the supplied interchange exponent and $\gamma = 1/(2-\tau)$ for the aggregate rectangle exponent. All coordinates below are address indices, rather than fields appended to every record. The tape volume counts their complete Cartesian array.

D.1 Weighted two-bank tile mask

For an integer pair (x, w), define the four-update word $P(a, b, s)$ by the chronological operations

$$\begin{aligned} x \mathrel{+}= [w \equiv b \pmod 2], \quad w \mathrel{+}= s[x \equiv a \pmod 2], \\ x \mathrel{-}= [w \equiv b \pmod 2], \quad w \mathrel{-}= s[x \equiv a \pmod 2]. \end{aligned}$$

The parity tests are evaluated at that step. Set $p = P(1, 1, +1)$ and $q = P(1, 0, -1)$. A direct check on the four parity tiles gives $p = (B\ C\ D)$ and $q = (A\ B\ D)$, with $A = (0,0), B = (1,0), C = (0,1), D = (1,1)$. Thus the chronological word p^{-1} followed by q is

$$(x, w) \longmapsto (2\lfloor x/2 \rfloor + (1 - (x \bmod 2)), w).$$

For a frozen control bit g, mask every one of these eight elementary updates by g. The resulting word is identity for $g = 0$ and the displayed bit flip for $g = 1$. In particular it restores the whole integer w, not merely its parity. Each of the eight increments is an ordinary integer update.

Now take u digit pairs with even radices n_i, m_i, all at least L^3, and ordinary numeric orders

$$X = \sum_{i=0}^{u-1} x_i A_i, \quad A_i = \prod_{j<i} n_j; \qquad W = \sum_{i=0}^{u-1} w_i B_i, \quad B_i = \prod_{j<i} m_j.$$

Each g_i is computed solely from frozen address controls disjoint from X, W and the dock below; it cannot depend on a bank while that bank is being updated. In one X-update of the eight-step word, let $\delta_i \in \{-1, 0, 1\}$ be its local signed increment, determined only by $w_i \bmod 2$, g_i, and the public word position. Execute the bank update by the single weighted integer increment

$$\Delta_X = \sum_i A_i \delta_i.$$

For a W-update, define ϵ_i from $x_i \bmod 2$ and set

$$\Delta_W = \sum_i B_i \epsilon_i.$$

The first offset is independent of X on each fixed (W, controls) fiber; the second is independent of W on each fixed (X, controls) fiber. Hence each update is a cyclic line rotation of its target numeric bank, not a gather of its component digits.

To implement it in the fixed-tape model, reserve a separate binary dock Z of width $c_0 = \max(a, b)$, where $a = \lceil \log_2 \prod_i n_i \rceil$ and $b = \lceil \log_2 \prod_i m_i \rceil$. Pad each target range once to 2^a or 2^b (factor below two), retaining its true validity predicate. For a selected native prefix, its numerical

range is a genuine Cartesian factor, not low binary bits of the flattened whole bank. If it is the high factor $[N_u][N_{\mathrm{rest}}]$, append entire opposite-factor rows; if it is the low factor $[N_{\mathrm{rest}}][N_u]$, append the invalid values in every low-factor block. In both cases the pad is below twice the original volume and costs one sequential scan with amortized counters. All other coordinates are opaque spectators. Only this selected factor is exchanged through the actual gap. For an X-update, interchange its a-bit field with an a-bit slice of Z using the supplied equal-chunk theorem. On the valid prefix $0 \le X < \prod n_i$, rotate modulo the true range by Δ_X; copy the invalid tail pointwise. If W is invalid, use offset zero on the valid X prefix. Interchange back. The W-update is symmetric, forcing zero offset when X is invalid. Every exchange restores arbitrary dirty dock contents. Consequently every privately padded state is fixed pointwise by each elementary update and may be cropped only after the completed word. No unequal-side transpose across an arbitrary gap is presumed. Validity follows the *logical target*: after docking X, test its value in the dock, not the old physical X position, which now contains arbitrary dock data. The other bank remains in place. The internal numerical pads of each SWAP call have already been cropped when that call returns, before this valid-prefix rotation.

Put $N_X = \prod n_i$, $N_W = \prod m_i$. Suppressing Cartesian spectators, the original native domain is $\Omega_0 = [N_X] \times [N_W]$ and the privately padded domain is $\widetilde{\Omega} = [2^a] \times [2^b]$. Every complete docked update fixes $\mathcal{I} = \widetilde{\Omega} \setminus \Omega_0$ pointwise. For the concrete schedule below keep these private pads through the repair, then crop them after the exact mask is complete.

On Ω_0, define the good set G_0 by quotient margins

$$10 \le \lfloor x_i/2 \rfloor \le n_i/2 - 11, \qquad 10 \le \lfloor w_i/2 \rfloor \le m_i/2 - 11 \quad \text{for all } i.$$

Each local integer moves by at most four during the eight updates, so on this set no digit overflows and the weighted updates equal the independent tile words. The ideal completed map toggles only x_i's endpoint bit; it preserves every quotient label and fixes W. It therefore preserves G_0 and $\mathcal{B} = \Omega_0 \setminus G_0$ separately. Extend the ideal map as identity on $\mathcal{I}$. The packed word P agrees with that extension on $G_0 \cup \mathcal{I}$, and both are permutations of $\widetilde{\Omega}$. Hence

$$P(\mathcal{B}) = \widetilde{\Omega} \setminus P(G_0 \cup \mathcal{I}) = \widetilde{\Omega} \setminus T(G_0 \cup \mathcal{I}) = T(\mathcal{B}) = \mathcal{B}.$$

The union bound gives density on the *original native domain*

$$\delta = \frac{|\mathcal{B}|}{|\Omega_0|} \le 40 \sum_i (n_i^{-1} + m_i^{-1}) = O(u/L^3).$$

The repair marker is explicitly

$$[X < N_X] \wedge [W < N_W] \wedge [\text{a quotient margin fails}].$$

It is false on $\mathcal{I}$, even if private padding occupies a constant fraction of $\widetilde{\Omega}$. Do not decode an invalid numeric bank as a wrapped native digit tuple. Extract only marked records at their current padded output addresses y. Recover their sources $x = P^{-1}(y)$ by reversing the eight packed updates, then compute the ideal endpoint destination $T(x)$. These are public address calculations with frozen controls; no old key is assumed to accompany the payload. Keys are destination ordinals in the same current padded layout. Stably sort these destination keys and refill the same bad slots. This corrects every exceptional address and discards no payload. The repair costs $O(\delta V L + M_{\mathrm{rec}} L^C)$ for a fixed C, where M_{rec} is the native record count before these private bank pads. The padded record count and bit volume are each below four times their native values, so full marking/refill scans cost $O(V)$. For clarity, with $K \le \delta M_{\mathrm{rec}}$ bad records, use stable binary radix sorting on the $b = O(L)$ key bits. In each pass copy one fixed-width key/payload item to a staging tape, remember the selected bit, rewind, and copy to the appropriate zero/one bucket. Concatenate the buckets and rewind for the next pass. A fixed number of tapes and $O(K(R_{\mathrm{op}} + b))$ steps suffice per pass, including all item and file returns.

Counter updates are amortized over the scanned length, and resetting a counter costs at most the following item or block length. The bound is therefore $O(KL(R_{\mathrm{op}} + L))$, plus explicitly charged polynomial address work and full-array marking/refill scans. At refill, consume the next sorted item exactly at a marked bad slot; $P(\mathcal{B}) = T(\mathcal{B}) = \mathcal{B}$ proves the order without a full-key comparison at every good record. The validity-and-margin predicate above marks $\mathcal{B}$ directly, so no extra mask lane is added to the retained field codec. The exponential opaque width below absorbs the key-only and metadata terms. The tile word uses a constant number of dock interchanges and rotations, hence costs $O(V(u \log R)^{\tau} + V)$. In the application below, the bundled opaque record width is exponential in L, so it absorbs $M_{\mathrm{rec}} L^{C}$; also $\delta L = O(u/L^2) = O(1)$. The resulting mask and its inverse cost

$$O(V(u \log R)^{\tau} + V).$$

After refill the padded map is exactly T on Ω_0 and identity on $\mathcal{I}$, with the dock restored. Crop only the private X, W numeric tails now, returning the original native ranges before any enclosing phase child. A caller's Fourier dummy p_i in a native range $n_i = p_i + 1$ is *inside* that range and is not such a tail; it remains part of the native kernel. Role-row, suffix and coefficient bit pads also retain their separate outer crop boundaries below.

D.2 Disjoint banks and available opaque lanes

Write $x_i = 2q_i + b_i$. The desired local matrix is H on $\{2q_i, 2q_i + 1\}$ for every q_i; hence it is $H^{\otimes a}$ on the endpoint bits at fixed quotient labels. This is an identity on fibers of the existing native array, not an uncharged regrouping.

Put $\ell_R = \lceil \log_2 R \rceil$ and

$$h = \min\{\lfloor a/16 \rfloor, \lfloor L/(40\ell_R) \rfloor\}.$$

For each selected block of at most $2h$ axes, one aggregate rectangle move places its native interval first. Take the selected endpoint planes as X, Y, and three disjoint h-axis banks from the other axes as Q, Z_0, W. The remaining axes form a spectator suffix. The jth digit is $2X_j + Y_j$; the two banks need not be physically interleaved. The banks Q, Z_0 are dirty selectors, and W is the separate tile auxiliary. In a shear updating any of these banks, its controls exclude the target, W and the binary dock. In particular the source-value toggle of Q reads X, Y, the toggle of Z_0 reads X, Y, Q, and a target transposition reads the other endpoint plane and Z_0. Each completed echo restores both dirty selectors.

Pad the whole spectator suffix once to its next binary range, with factor below two. Its index width s satisfies $s \geq 7L/8 - O(1)$, because $5h \log_2 R \leq L/8$. For the fixed final body of Lemma 3, including its full finite phase cover, reserve

$$r = \left\lceil \log_2 B^{\lceil \log_N (h+2) \rceil} \right\rceil = O_N(\log(h+2))$$

high suffix bits for independent numeric rows. Pad that row range once to a multiple of $B^{\lceil \log_N (h+2) \rceil}$; the factor is below two. Reserve the next $c_0 = \max_G \lceil \log_2 |G| \rceil \leq h \log_2 R + 1$ bits as a dock for $G \in \{X, Y, Q, Z_0, W\}$. The low suffix and original payload together have opaque width

$$R_{\mathrm{op}} = 2^{s-r-c_0} D \geq 2^{4L/5} D$$

above an effective fixed cutoff. Move only the r high row bits to the front, at $O(Vr)$ cost, once per completed block. After the row pad the numeric range need not be binary; its $r+1$-bit counter is metadata, not a further borrowed coordinate. Crop the numeric row pad before undoing these digit moves, then crop the whole suffix only after the completed block has fixed it.

Here and below an aggregate rectangle is padded as a whole, exchanged by Lemma 4, and cropped after its full endpoint. The volume factor is below four. No individual native axis is padded to a power of two. For a short selected block use the same available banks at its smaller width. Since $h \leq a/16$, its required auxiliary axes are disjoint from the selected ones.

D.3 Address generation and exact sparse repair

The instruction descriptors consist of fixed body-slot numbers, active rank lists, disjoint matchings, group placements, and a list of $O_N(\log h)$ inherited mask predicates. Decode the five bank indices by schoolbook division on $O(L)$-bit integers. Store rank lists in $O(L)$ records of $O(L)$ bits; a lookup is a sequential scan, not unit-cost indexing. A fixed-slot row operation pairs ranks $(i-1)f+j$ and $(k-1)f+j$ for $1 \le j \le f$. Interval placements are obtained by tracing their cycles and using two cycle reflections. The loops have bounds $O_N(L)$ or $O_N(L^2)$ and use binary arithmetic of polynomial cost. Thus one fixed polynomial $C_N L^c$ bounds the generation of any current key, selector, inverse packed update and source scalar. No expanding Boolean expression is substituted for the list of inherited predicates.

The width R_{op} dominates this polynomial beyond an effective fixed cutoff, even when $D = 1$. With K exceptional opaque records and $b = O(L)$ key bits, a staged stable radix sort costs $O(Kb(R_{\mathrm{op}} + b))$, plus $O(V)$ full marking/refill scans and $M_{\mathrm{rec}} C_N L^c$ metadata. The tile bad fraction is $O(u/L^3)$, so its payload repair is $O(Vu/L^2) = O(V)$. The active-prefix bad fraction below is at most $(L+2)^{-128}$ by increasing its fixed prefix constant. Its full-payload repair is also $O(V)$. Counter resets, bucket rewinds, erasure and head returns are included in these scans. Repair keys are computed from output addresses by the inverse packed map, never by an unrecorded input pointer.

D.4 Restored selectors and active-rank permutations

An A-digit has states $0, 1, 2$ active and state 3 idle. Represent $d_i = 2X_i + Y_i$. For any active mask S, $A^{\otimes h}$ restricts to $C^{\otimes |S|}$ on the active coordinates and identity on idle coordinates. All 4^h addresses remain in the stream; no sector is gathered out.

The mask-routing identities use dirty address fields. Let T_Z apply one fixed transposition of active symbols where dirty selector bit $Z_i = 1$. It preserves the active mask S, but arbitrary dirty Z can make it change paired sources as well as targets. Source-value predicates are recomputed only after the completed $T_{J \wedge Q}$ word below, which changes matched targets only. For a mask-only selector $J(S)$ the chronological word

$$Z \oplus\!= J(S), \quad T_Z, \quad Z \oplus\!= J(S), \quad T_Z \tag{18}$$

restores arbitrary initial Z and equals T_J. With a second dirty field Q, use $J(S) \wedge Q$ in (18). If $F_a(d)_i$ says the paired source of target i currently has trit a, then the completed $T_{J \wedge Q}$ changes targets only and preserves all paired sources. Consequently

$$Q \oplus\!= F_a(d), \quad T_{J \wedge Q}, \quad Q \oplus\!= F_a(d), \quad T_{J \wedge Q} \tag{19}$$

restores arbitrary Q, Z and implements the target transposition conditioned on source value a. The source/target matching must be disjoint; this is what makes the second F_a equal to the first at the completed inner-operation boundary. A third dirty field is reserved for the weighted tile auxiliary. The supplied weighted-mask lemma implements each frozen XOR in (18)–(19) with its separate auxiliary restored; its exact sparse boundary repair and interchange cost are not reproved here.

The two active transpositions have explicit bit circuits

$$(0\ 1):\ Y \oplus\!= Z \wedge \neg X, \qquad (1\ 2):\ X \oplus\!= Z \wedge Y; \quad Y \oplus\!= Z \wedge X; \quad X \oplus\!= Z \wedge Y. \tag{20}$$

They are constant sequences of frozen-control XOR shears. Applying the two transpositions in chronological order (1 2) then (0 1) adds one to an active trit; reverse order subtracts one. Conditioning on source trit and composing these shears realizes a matching SUM

$$\mathrm{target}_i \leftarrow \mathrm{target}_i + \mathrm{source}_i \pmod 3. \tag{21}$$

Three SUMs plus a negation implement a coordinate swap. All dirty selectors and the tile auxiliary return to their arbitrary incoming values after each completed operation.

For a node acting on the first e active ranks, restrict controls and endpoints to the first

$$u = \min\{h, \lceil C_0(e + \log(L + 2))\rceil\}$$

physical positions, where $L \geq h$ is the ambient address-width bound in the native implementation. If fewer than e active positions occur there, define the operation to be identity; otherwise all required ranks are present. Restrict every raw T_Z transposition and both selector-bank toggles to this prefix, not just the completed selector J. Insert inherited eligibility into $J(S)$; it is not a frozen control on an inner bit shear whose target can change S temporarily. Individual inner shears still read only the other bit plane and the selector bank. Their completed transposition preserves S. The comparison with the full intended route is made only after a *complete logical* mask-selected transposition, source-conditioned operation (19), SUM or recursive basis placement on the first e active ranks, with all its dirty selectors restored. Write this prefix implementation as P_u and its intended permutation as T. On a good eligible prefix all intended endpoints are inside it, so $P_u = T$. On a bad or ineligible prefix the selector J is zero: the two raw transpositions in (18) cancel, each inner operation is identity, and the two source-predicate toggles in (19) cancel too. Thus the completed P_u is identity there. These statements do not require raw dirty T_Z to preserve paired sources.

One must not repair an internal truncated T_Z to its full-h version: with $h = 2, u = e = 1$, $d = (0, 0)$ and $Z = (0, 1)$, the prefix is good but full T_Z for (0 1) changes the second digit and truncated T_Z does not. This toy example embeds for any actual $u < h$ by putting the dirty selector immediately after the prefix and using active digits throughout the prefix. That off-prefix action is intended to cancel in the outer word. Nor is the prefix bad set invariant after an inner bit shear: the first shear of (1 2) sends idle $(X, Y) = (1, 1)$ to active $(0, 1)$. Its tile repair uses a different invariant, as proved in the tile construction above. The separate top-level digit-group placement can reach ranks beyond e; it retains its full-h root-level charge and is not justified by this width-e prefix argument. In the uniform $[4]^h$ address space the prefix active count is binomial with mean $3u/4$. For $u \geq 2e$ and $u < h$, a Chernoff bound gives

$$\Pr\{B_u < e\} \leq e^{-u/24}. \tag{22}$$

For $u = h$, eligibility $|S| \geq e$ leaves no exceptional eligible mask. Both T and the completed logical P_u preserve S, so the exceptional set $E_{\text{inherited}}(S) \wedge [|S| \geq e] \wedge [B_u < e]$ is invariant. Its whole-universe fraction is bounded by the unconditional binomial estimate; no conditional distribution given eligibility is assumed. At that completed boundary, sorting the exceptional records by $T(P_u^{-1}(y))$ and refilling their slots repairs it exactly; the maps agree everywhere else. No sector is dropped. The full role recursion always stores the complete $[4]^h$ address universe, and its row split depends only on the independent role field, never on S. Thus every child retains the same mask distribution and off-eligibility maps are identity. Consequently, on every sector S, the role program applies $C^{\otimes|S|}$ to its active digits and identity to its idle digits; the direct sum over all masks is exactly $A^{\otimes h}$.

The prefix correction above belongs to the completed logical routing permutation, including T and T^{-1} below, after all its source conditioning and before a linear combination forms a phase. An internal $T_{J\wedge Q}$ is not separately corrected when its surrounding source predicate was prefix-truncated too. It never sorts the output of the non-permutation matrix C.

At selected width one, let T cyclically permute that eligible active trit and be identity on ineligible masks. The mask-only transpositions above implement T and T^{-1}. Three temporary array copies and a pointwise linear combination compute $I + \eta^2(T + T^{-1})$, which is C on the selected trit. Off eligibility the two identical terms cancel, leaving the identity. The copies, returns and cleanup add $O(V)$ at this base call. Norm two uses its inverse via the fixed coefficient wrapper described below.

D.5 The paired phase on the full address universe

For e paired coordinates, keep the first $2e$ active trits in two ordered halves. Every child inherits its parent's mask predicate; its domain test $|S| \geq 2e$ is conjoined with that predicate. A prefix failure is repaired at the completed logical routing boundary as above, rather than permanently disabling the ideal phase.

The rank-one kernel has an explicit nine-term formula. Summing the inverse Fourier expression first over one frequency gives

$$K_{I_1} = \sum_{a,b \in \mathbb{F}_3} \eta^{-ab} T_{(a,b)}, \qquad (T_{(a,b)} f)(x,y) = f(x+a, y+b).$$

Indeed $\sum_\zeta \eta^{\zeta(\xi-y)}$ is one for $\xi = y$ and zero otherwise in characteristic two. The coefficients sum to one. Use nine owned copies, the completed selected translations and a pointwise linear combination. Off the inherited gate every translation is identity, so the same formula is identity there. The copies, returns and erasure cost $O(V)$.

Freeze the final body with $a_* = P/B < N^z$ for a fixed $z < \tau$. Its charts are finite products of ternary row additions, negations and swaps, lifted by the column identity. The restored selector permutations above implement these exact disjoint-slot matchings. For $e = Nf + r$, $0 \leq r < N$, first align the Nf coordinates in each half, keeping both r-coordinate remainders together. This is a permutation of the original first $2e$ active ranks, implemented by two reflections of its cycles. Execute the body on the Nf pairs, place and process the r remaining pairs by bounded direct kernels, and undo the alignment.

The independent row stock was prepared once. Each child consumes one factor B and receives exactly V/B bits, including every mask, spectator, codec lane and retained root pad. Each of the P queries, including owned queries, is charged on that volume. Owned initialization, scalar maps and cleanup add $O_N(V)$ at the node. Before a child, push only the other current sibling streams onto a parking stack and copy its active stream. On return pop those siblings from right to left into their saved intervals. The parent boundary is marked; no descendant head crosses into a parked ancestor. A fixed set of input/output, role, owned, padding/repair and stack tapes suffices. Each of the fixed P push/pop operations costs $O_N(V)$ at its parent, not a multiplier on descendant time.

Consequently the time per active input bit satisfies, for one fixed polynomial Q in $\log(L+2)$,

$$F(e;L) \leq a_* F(\lfloor e/N \rfloor; L) + Q(L)((e + \log(L+2))^\tau + Q(L)).$$

At fixed bounded e use the direct kernels. Unrolling the e-dependent toll has ratio $a_*/N^\tau < 1$. Width-independent tolls total $O((e+1)^z \operatorname{polylog} L)$, with an extra logarithm if $a_* = 1$. Thus

$$F(e;L) = O((e+1)^\tau \operatorname{polylog} L).$$

No child repeats root layout, padding or lane encoding, and each temporary selected-factor pad is cropped before entering a child.

On four active trits, the chart J in Lemma 5 gives $U_{JT} C^{\otimes 4} U_{JT}^{-1} = K_{I_2}$. For each mask with b active trits, write $\lfloor b/4 \rfloor$ in base N. Its digits specify at most $N-1$ consecutive groups of $4N^j$ active ranks per level. A group placement is two cycle reflections; it costs $O(V(h \log R)^\tau \operatorname{polylog} L)$. Apply the chart, one paired rank-$2N^j$ job, its inverse chart and the inverse placement. Process the at most three remaining trits by direct C kernels. There are $O_N(\log(h+2))$ levels and the geometric sum of paired costs is $O(V(h+1)^\tau \operatorname{polylog} L)$. All grouping predicates depend only on the mask, so their description does not enumerate its exponentially many values. The completed map on every mask is $C^{\otimes b}$ on its active trits and identity on its idle trits. The direct sum is exactly $(C \oplus [1])^{\otimes h}$.

D.6 Completion of the binary endpoint tensor

Appendix E gives a fixed word of 384 forward phase letters and 392 monomial letters over $\mathbb{F}_{16}$ whose matrix is $H \otimes H$. Tensor its letters across the selected h pairs, rather than running a separate word per axis. Every monomial permutation on four states is affine on two binary bits and uses the weighted masks. If $M_{\pi(d),d} = m_d$, generate $\prod_j m_{d_j}$ at the source address, apply it and then route by π; the source-indexed scalar list is not reused at the output address. Its generator has the polynomial cost paid above.

Encode the D payload bits once into groups of four binary coefficient planes, with at most three zero bits of codec padding per record. The volume factor is at most four and is paid only at block entry. The four actual coefficient planes and all borrowed rows, docks and opaque lanes survive the complete word. The completed matrix is binary and fixes the borrowed coordinates, so zero codec padding returns to zero and may then be cropped. The high row pad crops before undoing its bit moves; the suffix pad crops afterwards. Private mask pads cropped at their earlier completed boundaries do not propagate into the phase recurrence. All crops and returns have their charged scans.

Since $a \le L/(3\log_2 L)$, blocks of at most $2h$ cover the axes in $\operatorname{polylog} L$ passes: either $h \ge a/32$ or $h \ge L/(80\ell_R)$. The finite remaining range has a direct branch. Aggregate layouts cost $O(VL^\gamma)$ per block; phase and monomial work costs $O(VL^\tau \operatorname{polylog} L)$. Because $\gamma > \tau$, the total is $O(VL^\gamma \operatorname{polylog} L)$, and in particular the bound stated in Lemma 5. The completed ordinary bit operator is symmetric. The same procedure computes its literal bit transpose; two applications compute its inverse because $H^3 = I$.

E A positive finite-field word for the binary endpoint tensor

The address field $\mathbb{F}_3$ above is not the phase field. For the constant word use

$$E = \mathbb{F}_2[t]/(t^4 + t + 1), \qquad \eta = t^5 = t^2 + t.$$

The quartic has no root in $\mathbb{F}_2$ and remainder 1 modulo the only irreducible quadratic $t^2 + t + 1$, so it is irreducible. Thus $E = \mathbb{F}_{16}$, it contains $\mathbb{F}_4 = \mathbb{F}_2(\eta)$, and $\eta^2 + \eta + 1 = 0$. The four-state phase letter is

$$A = (\eta I_3 + \eta^2 J_3) \oplus [1] = C \oplus [1].$$

It acts separately on the two $\mathbb{F}_4$ coefficient planes in $E = \mathbb{F}_4 \oplus t\mathbb{F}_4$. Its inverse is A^2. The inverse-phase wrapper on these planes is

$$\widehat{\sigma}(a + tb) = a^2 + tb^2 \quad (a, b \in \mathbb{F}_4), \tag{23}$$

which applies the $\mathbb{F}_4$ Frobenius to each plane and fixes the codec basis element t. It is $\mathbb{F}_2$-linear and is not ordinary $\mathbb{F}_{16}$ squaring. In the polynomial basis $(1, t, t^2, t^3)$ it is the explicit map

$$(c_0, c_1, c_2, c_3) \longmapsto (c_0 + c_2 + c_3,\ c_1 + c_3,\ c_2,\ c_3).$$

Indeed $a_0 = c_0$, $a_1 = c_2 + c_3$, $b_0 = c_1 + c_2 + c_3$ and $b_1 = c_3$ in $a = a_0 + a_1\eta$, $b = b_0 + b_1\eta$. Three in-place XORs implement it, and repeating them gives the identity. For $\lambda \in \mathbb{F}_4$, $\widehat{\sigma}(\lambda z) = \lambda^2 \widehat{\sigma}(z)$. Consequently coefficientwise conjugation gives

$$\widehat{\sigma} C \widehat{\sigma} = \eta^2 I_3 + \eta J_3 = C^2 = C^{-1}, \qquad \widehat{\sigma}(C^{\otimes f})\widehat{\sigma} = (C^{-1})^{\otimes f}.$$

This holds on arbitrary E-valued inputs. The two passes cancel on idle or ineligible coordinates; either pass alone can change their coefficient values. Both passes act within the existing four-bit lanes, in $O(V)$ time around one child. Ordinary E squaring used on both sides would fail even on an idle value t, sending it to $t^4 = t + 1$ instead of t. The semilinearity assertion is only for $\mathbb{F}_4$ scalars, not arbitrary E scalars.

Here is the explicit word construction. Put $a = \eta^2$ and $b = \eta$; use column vectors and rightmost-first composition. Then

$$A^3 = I, \quad A^{-1} = A^2, \quad u = (1, a, a, 0)^T, \quad v = (1, b, b, 0)^T, \quad P = AE_{11}A^{-1} = uv^T,$$

with $v^T u = 1$, hence $P^2 = P$. Let

$$D = \operatorname{diag}(t, 1, t+1, 1), \quad Q = DPD^{-1} = x_0 y_0^T,$$

where $x_0 = (t, a, a(t+1), 0)^T$ and $y_0 = (t^{-1}, b, b/(t+1), 0)^T$. Direct products give

$$v^T x_0 = 0, \qquad y_0^T u = \kappa = \frac{t^2+t+1}{t(t+1)} \neq 0, \qquad PQ = 0, \quad QP = \kappa x_0 v^T. \tag{24}$$

For idempotents P, Q with $PQ = 0$,

$$[I + cP, I + dQ] = I + \frac{cd}{1+c} QP. \tag{25}$$

This follows by using $(I+cP)^{-1} = I + c(1+c)^{-1}P$ and $QPQ = 0$. With $c = d = \eta^2$, $cd/(1+c) = 1$. Thus

$$\begin{aligned} G &= A \operatorname{diag}(\eta, 1, 1, 1) A^{-1} = I + \eta^2 P, \\ H &= DGD^{-1} = I + \eta^2 Q, \\ T_0 &= GHG^{-1}H^{-1} = I + \kappa x_0 v^T. \end{aligned} \tag{26}$$

Each of G, H, G^{-1}, H^{-1} uses three forward A letters after replacing inverse A by A^2, so T_0 uses 12.

The remaining conjugations isolate an elementary matrix. Set

$$B = \operatorname{diag}(1, \eta, \eta, 1), \quad x = (t, 1, t+1, 0)^T, \quad w = (1, 1, 1, 0)^T.$$

Then $T_1 = BT_0B^{-1} = I + \kappa x w^T$ (12 A letters). Let P_{12} swap coordinates 1 and 2. Since w is fixed and $w^T P_{12} x = 0$,

$$T_2 = T_1 P_{12} T_1 P_{12}^{-1} = I + \kappa(t+1)(e_1 + e_2) w^T \tag{27}$$

uses 24 A letters. With $D_3 = \operatorname{diag}(1, 1, \eta, 1)$,

$$T_3 = T_2 D_3 T_2 D_3^{-1} = I + \kappa(t+1)\eta(e_1 + e_2) e_3^T \tag{28}$$

uses 48; the two cross pairings vanish and the row sum is $(1 + \eta^{-1})e_3^T = \eta e_3^T$. With $D_1 = \operatorname{diag}(\eta, 1, 1, 1)$,

$$T_4 = T_3 D_1 T_3 D_1^{-1} = I + \kappa(t+1) E_{13} \tag{29}$$

uses 96; the cross pairings vanish and the column factor is $1 + \eta = \eta^2$. Put $\mu = \kappa(t+1) \neq 0$; then

$$T_5 = \operatorname{diag}(\mu^{-1}, 1, 1, 1) T_4 \operatorname{diag}(\mu, 1, 1, 1) = I + E_{13}. \tag{30}$$

All conjugating factors are invertible monomial matrices. Coordinate permutations produce every $U_{ij} = I + E_{ij}$ with the same 96 forward A letters. This is an explicit formula, not an uncomputed word search.

Finally write $H_2 = US$, with $U = I + E_{12}$ and S the two-coordinate swap. In ordered coordinates $(00, 01, 10, 11)$, put

$$P_{\text{first}} = (1\ 3)(2\ 4), \qquad P_{\text{second}} = (1\ 2)(3\ 4).$$

Then, using the four explicit elementary-matrix words,

$$H_2 \otimes H_2 = U_{13} U_{24} P_{\text{first}} U_{12} U_{34} P_{\text{second}}. \tag{31}$$

There are four U_{ij} factors, hence $4 \cdot 96 = 384$ forward A letters; the remaining factors expand to 392 invertible monomial letters, for 776 total. The completed matrix has entries in $\mathbb{F}_2$. As an $\mathbb{F}_2$-linear map on the four binary coordinates of an E-symbol, it applies $H_2 \otimes H_2$ independently to each coordinate plane. Thus a binary 0/1 embedding returns to 0/1, and any zero lane used for padding returns to zero at the completed word. An arbitrary E-valued input need not return to the scalar subfield. Intermediate E lanes must not be cropped. When four arbitrary payload bits are packed into one E symbol, all four coefficient planes remain data; only extra zero bits from the final partial group are removed. No projection $E \to \mathbb{F}_2$ is applied to such a packed symbol.

Pairing $2h$ identical native endpoint axes and tensoring this fixed word reduces their algebraic action to $A^{\otimes h}$ letters and fixed monomial letters. The permutation part of each monomial is in $\mathrm{AGL}(2, \mathbb{F}_2) = S_4$: its NOT/CNOT address shears and the product of its h diagonal E scalars are charged by the separately supplied native mask/routing and record-scan interfaces. The active-sector role construction reduces the $A^{\otimes h}$ letters. The native large-even-radix endpoint appendix supplies their tape price. No binary 2^k-axis H2 bound is asserted here; that requires a distinct binary mask-routing argument and is not needed for the manuscript's native endpoint interface. For precision, write a monomial as $M_{\pi(d),d} = m_d$, with $m_d \in E^*$. Apply the coefficient $\prod_i m_{d_i}$ at its current *source* address, then route by $\pi^{\otimes h}$. Equivalently an output-address scan would use $\prod_i m_{\pi^{-1}(e_i)}$. Using the source-indexed list unchanged after routing is generally incorrect. Tensoring the fixed word is letterwise: $(L_1 \cdots L_{776})^{\otimes h} = L_1^{\otimes h} \cdots L_{776}^{\otimes h}$. Thus there are 384 tensor phase calls, not 384 separate calls per axis.

F Formal convolution, linear replay and coefficient arithmetic

This appendix makes the local characteristic-two substitution explicit. It does not import the complex convolution service from [5].

Newton symmetry over a ring. Let P have as column k the monomial coefficients of $\prod_{a<k}(Z - u^a)$. It is unit upper triangular, and evaluation gives $F_p(u)P = N$. With $D = \mathrm{diag}(N_{kk})$ and $L = ND^{-1}$, this is the unit-lower/diagonal/unit-upper factorization $F_p(u) = LDP^{-1}$. Such a factorization with invertible diagonal is unique over a commutative ring: comparing two makes a unit lower triangular matrix equal an upper triangular one, identifying the diagonal and then the two unit triangular factors. Since $F_p(u)$ is symmetric, its transposed factorization is another of that form. Therefore $P^{-1} = L^T$ and $F_p(u) = ND^{-1}N^T$. The pivot inverses are exactly the units proved in the main text; no field cancellation has been assumed.

A shared eight-H schedule. Write $D_a = \mathrm{diag}(a, 1)$, with $a = \eta$ or η^2. Direct multiplication in $\mathbb{F}_4$ gives

$$H^3 = I, \qquad HD_\eta H^2 D_{\eta^2} H = \mathrm{diag}(\eta^2, \eta)J,$$

and

$$W_4 := HD_\eta HD_{\eta^2} HD_\eta HD_{\eta^2} = I.$$

Thus $I = W_4 W_4$, J is its displayed four-H word with a leading unit diagonal followed by W_4, and $H = H^4 W_4$ all have eight H letters with unit diagonals in common slots. Identity diagonals fill missing gaps. The formulas include all identity pairs and never select a zero scalar. Coefficients may depend on a pair label because every intermediate letter preserves that label. Controlled coefficients are evaluated only at the completed K_T boundaries described in the main proof.

Formal convolution over a coefficient algebra. For comparison, Cantor–Kaltofen [9] give sharper classical size bounds and logarithmic arithmetic depth. Their sparse integer *stage* matrices compose to a formal bilinear identity; the composed matrices need not be sparse. We use

the following elementary triadic construction instead, so its descriptor generation and coefficient types can be checked directly.

Lemma 6 (Elementary formal convolution)**.** *Over any commutative $\mathbb{F}_2$-algebra A, ordinary multiplication of two length-v polynomials has a division-free formal bilinear circuit with size $O(v\log^8(2v))$, depth $O(\log(2v))$, binary fan-in, and deterministic polynomial-v wire-descriptor generation.*

Proof. Use schoolbook products below the fixed threshold $v_0 = 6^6$, with balanced output sums. Otherwise let b be the least power of three with $b^2 \geq v$, put $s = \lceil v/b \rceil$, and let t be the least power of three with $t \geq 2s-1$. Then

$$\sqrt{v} \leq b < 3\sqrt{v}, \quad s \leq b, \quad t \leq 3b, \quad t < 6s, \quad t \mid 3b.$$

Work formally in $R = A[Y]/(Y^{2b}+Y^b+1)$, free of rank $2b$ over A. The polynomial is Φ_{3b} reduced modulo two. Its roots over $\overline{\mathbb{F}}_2$ have exact order $3b$: they satisfy $Y^{3b}=1$, and none satisfies $Y^b = 1$ since the defining polynomial then equals one. Every proper divisor of $3b$ divides b. Therefore $Y^j - 1$ is coprime to this polynomial for $0 < j < 3b$. The corresponding Bezout identity extends from $\mathbb{F}_2$ to A, proving principality of Y and of $\zeta = Y^{3b/t}$ at order t. The inverse DFT uses ζ^{-1}; its normalization is one because t is odd.

Encode the input as $\sum_{i<s} A_i(Y)Z^i$, with $\deg A_i < b$, and similarly for the other operand. Take their t-point DFTs over R. At a frequency the value is *one* R element, represented by $2b$ coefficients from A; no unevaluated Z variable remains. Multiply two such values by one recursive ordinary length-$2b$ product over A, and reduce the degree-$\leq 4b-2$ result modulo $Y^{2b}+Y^b+1$. For $0 \leq r \leq 2b-2$, the high monomial reductions are

$$Y^{2b+r} = \begin{cases} Y^{b+r} + Y^r, & r < b, \\ Y^{r-b}, & b \leq r \leq 2b-2. \end{cases}$$

They cost $O(b)$ additions in constant depth.

The inverse DFT recovers $C_j(Y) = \sum_{i+k=j} A_i(Y)B_k(Y)$ without wrap in j, since $t \geq 2s-1$. Each true C_j has degree at most $2b-2$, below the quotient degree, so reduction has lost no coefficient information. Reconstruct the ordinary product as $\sum_j C_j(X)X^{bj}$: adjacent length-$2b$ blocks overlap, and their overlaps are added, not concatenated. Crop the known padded zero tail to length $2v-1$.

Multiplication by a root power in R pads a $2b$-coordinate vector to $3b$, rotates, and folds $(a_0,a_1,a_2) \mapsto (a_0+a_2, a_1+a_2)$. Each coordinate uses at most two inputs. A radix-three butterfly therefore has $O(b)$ binary additions and constant depth over A. Explicitly, at transform length $t = 3u$ split inputs as x_{3j+r}, $r = 0,1,2$. Compute their three length-u transforms $B_r(q)$ with root ζ^3, then print

$$\widehat{x}_{q+\ell u} = \sum_{r=0}^{2} \zeta^{r(q+\ell u)} B_r(q), \qquad 0 \leq q < u, \quad \ell = 0,1,2.$$

This recursively specifies every butterfly and twiddle index. The two forward and one inverse transforms, pointwise folds and reconstruction give, for an effective absolute constant C,

$$S(v) \leq tS(2b) + Ctb\log(2t), \qquad D(v) \leq D(2b) + C\log(2t).$$

These counts are in A gates, not unit-cost R operations. Moreover

$$2bt < 12bs \leq 12(v+b) \leq 24v, \qquad 2b < 6\sqrt{v} \leq v^{2/3} \quad (v \geq v_0).$$

Induction using $S(v) \leq Kv(\log_2 v)^8$ above the direct branch contracts the recursive term by

$$24(2/3)^8 = 6144/6561 < 1.$$

The $O(v \log v)$ toll is absorbed by increasing a fixed K; the finite direct branch fixes an effective base constant. This proves the size bound. The logarithms of child lengths contract by $2/3$, so the depth recurrence sums to $O(\log(2v))$.

All nonrecursive maps are the printed linear maps; recursive leaves multiply a left-operand form by a right-operand form. This proves a formal bilinear identity over A, without $x^2 = x$. For generation, compute b, s, t by integer loops, emit radix-three butterflies with rotation/fold source indices, and recursively emit the t children. Each descriptor has $O(\log v)$ address bits; source indices can be obtained by schoolbook arithmetic and sequential list scans in polynomial time. This finite recursive printer is uniform, with no root table, factorization, or coefficient-dependent branch. □

Here is a second explicit way to see that specialization leaves only linear gates. After fixing the kernel, attach to each gate its constant value c_g and its homogeneous degree-one part ℓ_g in the remaining input indeterminates. An input x_i has pair $(0, x_i)$, a fixed scalar a has $(a, 0)$, and the rules are

$$(c, \ell) + (d, m) = (c + d, \ell + m), \qquad (c, \ell)(d, m) = (cd, c\,m + d\,\ell).$$

The constants are prepared once. Only additions and multiplication by these fixed constants remain in the linear DAG. The rules are formal truncation modulo total degree at least two; they use no Boolean relation $x_i^2 = x_i$. Since fixed-kernel convolution is homogeneous linear, its output is exactly the degree-one part. The transformation changes size and depth by constant factors.

Bounded-degree copy trees, including copies of inputs and outputs, cost size proportional to the number of use edges. Replacing each gate's fan-out by its balanced tree adds at most $O(\log v)$ depth per original circuit layer. The deliberately loose linear-DAG bounds are therefore

$$g(v) = O(v \log^8(2v)), \qquad H(v) = O(\log^2(2v)).$$

All wire and scalar descriptors have polynomial-time generation in v and the explicit coefficient dimension. Kernel-specific constant propagation is charged once in that setup, with the actual coefficient words retained. This construction is independent of the faster multiplier being proposed.

Replay on arbitrary dirty coordinates. Let a linear DAG implement $M : A^e \to A^a$, and assign a distinct borrowed coordinate to each of its g gates. The physical source coordinates x, target coordinates y, and borrowed coordinates z are pairwise disjoint. Every output is a designated wire; any additions needed to form it and any input/output copy trees are included in the counted DAG. Thus the output-read edges also have bounded degree. In the notation below, F, D name these actual port reads, rather than uncosted dense matrix applications. Replace gate assignment by addition of its predecessor form into that coordinate, in topological order. For initial borrowed vector z this sweep has form $z \mapsto Uz + Tx$, where U is unit lower triangular. If the prescribed reads are $Fz + Dx$, zero initialization would compute the DAG, so $FT + D = M$. Sweep, add these reads to y, and undo the sweep. The net increment is $Mx + FUz$, while z is restored. Now sweep with every direct edge from x omitted, including edges to input-copy-tree roots, and add only the borrowed-coordinate read FUz to y, and undo again. In characteristic two the two FUz terms cancel. This realizes $(x, y, z) \mapsto (x, y + Mx, z)$ for every initial z, including coordinates correlated with current sources and targets.

At each DAG depth destinations are fresh gate coordinates and predecessors are earlier ones. Bounded fan-in and fan-out give a bounded-degree graph of shear edges. A constant edge coloring makes each color disjoint pairs. Apply its colors before advancing to the next depth; undo in reverse depth and color order. The output-port edges into the distinct y coordinates have the same constant-color bound. In the second read omit every direct x-to-output edge

as well. Zero-coefficient edges may be omitted throughout. The four sweeps and two output rounds thus use $O(H+1)$ shear layers, on exactly $e+a+g$ coordinates. No clean gate values or fresh zero coordinates are assumed. Other parent coordinates can be selected as borrowed gates, even if they contain data from an earlier chunk operation; the replay restores them exactly. A source/target alias is outside this contract.

Explicit rectangular displacement reconstruction.

Lemma 7 (Exact-width unit Toeplitz word)**.** *Over any commutative $\mathbb{F}_2$-algebra A, a unit lower triangular Toeplitz matrix T_v has a word on exactly its original v coordinates of $O(\log^{19}(2v))$ layers. Each layer consists of shears $(x,y)\mapsto(x,y+\mu x)$ on disjoint ordered pairs. No clean coordinate is required. Descriptors and coefficients can be prepared by polynomially many operations on indices, finite lists, and public coefficient-algebra elements.*

Proof. Write $(T_v)_{ij}=h_{i-j}$ for $i\ge j$, with $h_0=1$. Its reciprocal sequence is prepared by $g_0=1$ and $g_k=-\sum_{j=1}^{k}h_jg_{k-j}$; no coefficient inversion is needed. For $v\ge 2$ split at $s=\lfloor v/2\rfloor$ and put $r=v-s$. The matrix has blocks

$$T_v=\begin{pmatrix}T_s & 0\\ B & T_r\end{pmatrix}.$$

After applying the two diagonal children in place, a cross update $E=BT_s^{-1}$ gives this matrix exactly. Its entries are

$$E_{ij}=\sum_{\nu=j}^{s-1}h_{i-\nu}g_{\nu-j}.$$

For consecutive row/column chunks, the interior difference is

$$E_{ij}-E_{i-1,j-1}=-h_{i-s}g_{s-j}.$$

Let M be that rectangular chunk and let J_a,J_e be lower shifts. Its difference $\mathcal{D}=M-J_aMJ_e^T$ is the displayed outer product plus first-row and first-column corrections, hence the sum of three explicit outer products. More explicitly, let the chunk's absolute row and column indices start at $i_0\ge s$ and $j_0<s$, and use zero-based local indices. Set

$$v_i=-h_{i_0+i-s},\qquad w_j=g_{s-j_0-j},\qquad r_j=M_{0j}-v_0w_j,$$

and set $c_0=0$, $c_i=M_{i0}-v_iw_0$ for $i>0$. Then, with e_0 the first standard vector of the appropriate length,

$$\mathcal{D}=vw^T+e_0r^T+ce_0^T.$$

The interior follows from the displayed difference identity, the second term corrects the complete first row, and the third corrects the remaining first column without counting $(0,0)$ twice. Every entry is given by finite sums and products of the prepared Toeplitz and reciprocal coefficients. This uses no field rank test. Nilpotence gives $M=\sum_{l\ge0}J_a^l\mathcal{D}(J_e^T)^l$. For one term vw^T, its action on x is

$$b_l=\sum_{j=l}^{e-1}w_{j-l}x_j,\qquad c_i=\sum_{l=0}^{e-1}v_{i-l}b_l.$$

Here $v_k=0$ outside $0\le k<a$. For literal convolution windows put $W(Z)=\sum_{j<e}w_jZ^j$, $X^{\mathrm{rev}}(Z)=\sum_{j<e}x_jZ^{e-1-j}$ and $V(Z)=\sum_{i<a}v_iZ^i$. The polynomial with coefficients b_l is

$$\mathrm{rev}_e\big((WX^{\mathrm{rev}})\bmod Z^e\big),$$

and the output polynomial is its product with V, reduced modulo Z^a. Here the reversal at fixed padded width is

$$\mathrm{rev}_e\, P = Z^{e-1}P(Z^{-1}).$$

The two ordinary products have output lengths at most $2e-1$ and $a+e-1$, respectively. If $a, e \le b$, both are at most $2b-1$. Their padding zeros are formal constants; they introduce no extra runtime source coordinates. Reversal and selection specify DAG ports. The three displacement terms therefore use six fixed-kernel convolution circuits and their sum, with total gate bound $O(b\log^8(2b))$ and depth $O(\log^2(2b))$ including all intermediate and copy coordinates.

At setup enumerate $b = 1, \dots, v$, partition both sides of the cross block into consecutive chunks of width at most b, and print the actual DAG for every pair, including ragged final chunks and its input/output copies. Choose the largest b for which $e+a+g \le v$ for every pair. For a uniform fixed C the count is $e+a+g \le 2b+Cb\log^8(2b)$. Thus $b_0 = \lfloor cv/\log^8(2v)\rfloor$ passes for sufficiently large v and a sufficiently small fixed $c > 0$; the chosen size is at least b_0. If none passes, use the direct triangular shear word, updating targets in descending order from their earlier source coordinates. This branch occurs only in a fixed finite range.

For each accepted pair its source chunk lies in $[0, s)$ and its target chunk in $[s, v)$, hence they are disjoint. Choose the g borrowed coordinates from their complement inside this same v-coordinate parent. Replay restores that complement before the next pair. The source half is unchanged by every completed cross update, so the sum of all chunk updates is exactly E. There are $O(\log^{16} v)$ chunk pairs, each with $O(\log^2 v)$ replay layers. The diagonal children use only their own disjoint intervals. Run their matching layers in parallel, padding the shorter word with identity layers. Their union is again a matching. Thus

$$d(v) \le \max(d(\lfloor v/2\rfloor), d(\lceil v/2\rceil)) + O(\log^{18}(2v)), \qquad d(v) = O(\log^{19}(2v)).$$

All coordinates belong to the original v-coordinate parent. Preparing and counting graphs, reciprocal coefficients, outer-product generators and edge colors uses only polynomial-size finite lists and polynomially many public coefficient operations. The search is setup for the public kernel, never a per-payload-record computation. □

The Newton Toeplitz sequence is $h_j = H_j^{-1}$, so $h_0 = 1$ as required by Lemma 7. Transposing a layer word reverses its layer order and each shear's source/target direction; this supplies the upper triangular word on the same coordinates at the same layer count. This transposition is on A-coordinate indices and retains each coefficient μ. It is a descriptor operation, not a physical array transpose. Consequently $F_p = ND^{-1}N^T$ uses two such Toeplitz words and a fixed number of unit-diagonal stages. Synchronize different prime axes by identity matching layers. With $p \le R_0$, preparing all their words and actual B-coefficient tables has bit cost polynomial in R_0, D', L_0, as in the coefficient setup below. Each matching layer is then replaced by the fixed local slot word in the main text; these abstract layer counts do not assign unit bit cost to growing scalars or tape movement.

The actual constant extension. Every irreducible factor of $\Phi_{\alpha'}$ has degree $r = \mathrm{ord}_{\alpha'}(2)$, so S is a product of copies of $\mathbb{F}_{2^r}$. A factor tensored with $\mathbb{F}_4$ splits into $\gcd(r,2)$ fields of degree $r_B = \mathrm{lcm}(r,2)$. Therefore the same Frobenius exponent in the main proof selects $\mu = 1$ in every field component of B, with $r_B \le 2D'$. Computing r by repeated multiplication by 2 modulo α' needs at most D' iterations and has polynomial small setup cost. Exponentiation for the selector needs $O(r_B)$ ring products. The fixed $\mathbb{F}_{16}$ codec inside the binary H supplier need not embed into B: the completed H acts identically on each of the $2D'$ binary lanes and is thus the required B-symbol map.

The central $\mathbb{F}_4$ in B and a fixed field codec used inside a binary routing/H kernel need not have the same intermediate bit encoding. The replacement is made only at a completed binary kernel: its target matrix is H, I or J on whole symbols and acts identically on every binary

lane. It therefore implements the required B-linear map in the caller's original representation. Outer multiplications by elements of B are inserted only after that boundary and always use the separately charged coefficient-ring arithmetic. Even a central $\mathbb{F}_4$ action on two concatenated S lanes can be implemented by a constant number of aligned word scans; no growing basis conversion or free bit transpose is assumed.

Actual growing-word arithmetic and unit setup.

Lemma 8 (Fixed monic remainder at any input width)**.** *Fix a monic binary polynomial f of degree $d \geq 1$ and a public input width $r \geq d$. After polynomial-r setup, the remainder modulo f of any binary polynomial of degree $< r$ costs $O(M_{\mathrm{old}}(r))$ bit steps on the fixed tape model, including all scans and returns.*

Proof. If $r = d$, the input is already reduced. Otherwise set $k = r - d > 0$. The reversal $f^* = X^d f(X^{-1})$ has constant coefficient one by monicity. Prepare $h = (f^*)^{-1} \bmod X^k$ once, for example by its coefficient recurrence with elementary polynomial-time tape arithmetic. For the dividend w put

$$w^* = X^{r-1} w(X^{-1}), \qquad q^* = (w^* h) \bmod X^k, \qquad q = X^{k-1} q^*(X^{-1}), \qquad R = w - qf.$$

All reversals have the displayed fixed padded widths. To prove the formula, write the unique monic division $w = Qf + R_0$ with $\deg Q < k$ and $\deg R_0 < d$. Reversal gives $w^* = Q^* f^* + X^{r-1} R_0(X^{-1})$; the second term is divisible by X^k. Multiplying by h modulo X^k therefore recovers Q^*, so $q = Q$ and $\deg R < d$.

The first product may truncate w^* to its first k coefficients. It has two inputs of length k; the other product has input lengths k and $d + 1$, each at most r. Thus two calls to the independent word supplier cost $O(M_{\mathrm{old}}(r))$, and all reversals, cropping, XOR and head returns add $O(r)$. There is no per-dividend Euclidean algorithm. The prepared inverse length is k, even when $d \ll r$. □

Put $d = D'$ and $f = \Phi_{\alpha'}$. Store a B symbol as the two consecutive d-bit S words in $a + b\eta$. If $P = ac$, $Q = be$, and $T = (a + b)(c + e)$ in S, then

$$(a + b\eta)(c + e\eta) = (P + Q) + (T + P)\eta.$$

Three binary polynomial products modulo fixed f, supplied by Lemma 15, with linear XOR scans, therefore suffice. Their degree-$\leq 2d - 2$ outputs use Lemma 8 at input width $2d - 1$. Its reciprocal is prepared at length $d - 1$ for $d \geq 2$; $d = 1$ is direct. This proves $O(M_{\mathrm{old}}(d))$ application time with no dense d-by-d matrix operation. The runtime word supplier is classical integer SS with carry-free binary packing; it is separate from the formal characteristic-two convolution circuit used to construct the local linear DAG above.

The involution $\eta \mapsto \eta + 1$ has norm $N(a + b\eta) = a^2 + ab + b^2 \in S$. For a unit $x = a + b\eta$, its norm is an S unit: it is a product of units in B, and an inverse in the free extension $B = S \oplus S\eta$ supplies an inverse in S by taking its constant coefficient. Thus

$$x^{-1} = ((a + b) + b\eta)\, N(x)^{-1}.$$

Polynomial extended Euclid with f suffices for the norm inverse, without factoring B. Such inverses are prepared once in polynomial time in the small dimensions. The selector exponent $2^{r_B} - 1$ also needs only $O(r_B)$ binary squaring/product steps, with $r_B \leq 2d$. For example, from $z = \mu + 1$ repeat $z \leftarrow z^2(\mu + 1)$ exactly $r_B - 1$ times. No possibly nonunit $\mu + 1$ is inverted. The one-time polynomial setup is separate from every runtime scalar application.

Scalar lists on fixed tapes. At a synchronized local slot, each axis has a public table $a_i(x_i) \in B^*$ in that slot's current coordinate order; the tensor diagonal is $\prod_i a_i(x_i)$. Pairing and coefficient choices are axis-local. Jointly controlled CRT coefficients use their different generator and are not covered by this factorization.

Let $M_j = \prod_{i \le j} n_i$, $M_0 = 1$, with axes in physical lexicographic order. Start with the one-word list [1]. For every partial product c on the current list, scan the entire next local table and print $ca_j(0), \dots, ca_j(n_j - 1)$. Rewind that local table, then advance to the next c. Its rewind costs the size of the child block just printed. Alternate two expansion tapes and erase the previous list at the end of each level. This keeps the produced list in the current native order and uses a fixed tape set.

Level j uses M_j growing-ring products and $O(M_j(d + L))$ copying, counters and returns. Here $2d \ge L$, and all $n_i \ge 2$, so

$$\sum_j M_j < 2M, \qquad \text{cost(generate and apply one tensor diagonal)} = O(M M_{\text{old}}(d)).$$

Apply the final list to the payload stream in lockstep and erase it. Prepare or select each local table once per axis/slot in the polynomial small setup, using one shared archive and one reusable active-table tape; there is no tape per axis or indexed archive lookup per payload record. Rebuild the list from the next slot's tables after completed relabeling, H and J calls. Transporting an old scalar list and presuming alignment is not part of the program.

G Controlled tensors and the ordinary coefficient reduction

Write $\tau = t$ for the equal-field interchange exponent and $\gamma = g = 1/(2 - t)$ for the native endpoint and aggregate rectangle exponent proved above. Constants are fixed before the input length varies. For native dimensions $M = \prod_i n_i$, set $L = \lceil \log_2(2M) \rceil$ and $V = MD$. The generic permutation tariff established below is

$$O(VL^\gamma \operatorname{polylog} L). \tag{32}$$

This appendix develops the outer construction of [3]; the arithmetic and movement costs use the suppliers proved in this paper.

G.1 A tensor perfect shuffle

Write $n_i = 2k_i$ with the first digit most significant. Define

$$S_i(k_i h_i + \ell_i) = 2\ell_i + h_i, \quad h_i \in \{0, 1\}, \quad 0 \le \ell_i < k_i.$$

Also define $d_M(X) = 2X \bmod (M - 1)$ for $X < M - 1$ and fix $M - 1$. Splitting the flat stream into its two halves and interleaving records implements d_M in $O(V)$ time.

Lemma 9 (Post-shuffle endpoint correction)**.** *The tensor $\bigotimes_i S_i$ and its inverse have cost*

$$O(VL^\gamma \operatorname{polylog} L)$$

by Lemmas 4 and 5.

Proof. For $x_i = k_i h_i + \ell_i$ and incoming carry $c \in \{0, 1\}$,

$$2x_i + c = n_i h_i + (2\ell_i + c).$$

The outgoing carry is exactly h_i, independent of c. End-around doubling therefore has output $y_i = 2\ell_i + h_{i+1}$, cyclically in i. Consequently

$$\bigotimes_i S_i = r \circ d_M, \qquad r(y)_i = 2\lfloor y_i/2 \rfloor + (y_{i-1} \bmod 2).$$

This rotates low endpoint bits *after* the flat shuffle; it does not require a cheap rotation of the original high-half bits.

Index the axes from zero. Reverse endpoint bits on $[0, t-1)$ and then on $[0, t)$. A bit at $i < t-1$ moves first to $t-2-i$ and then to $i+1$; the last bit moves to zero. This is precisely r. For a reversal $[a, b)$, a left-half interval $I = [c, d)$ pairs with $J = [a+b-d, a+b-c)$, in reverse local order. Partition the left half into such intervals of length $u \le h$, where

$$\ell_R = \lceil \log_2 R \rceil, \qquad h = \min\{\lfloor t/16 \rfloor, \lfloor L/(40\ell_R) \rfloor\}.$$

The complement of two selected intervals has at most three linear intervals. Since $t \ge 16h$, one has at least u axes; choose a contiguous u-axis tile auxiliary W there.

Place the two selected banks X, Y and W at the front in that order. Extract them successively from the remaining axis list: at each step exchange the whole preceding aggregate gap with the requested interval. At most three adjacent aggregate rectangle exchanges suffice. All unselected axes keep their relative order; reverse the same exchanges after the batch. Each exchange crops its two aggregate pads before the next one and costs $O(VL^\gamma)$. No endpoint gather is performed.

With $\pi(j) = u - 1 - j$, the chronological endpoint masks $Y_{\pi(j)} \oplus= X_j$, $X_j \oplus= Y_{\pi(j)}$, $Y_{\pi(j)} \oplus= X_j$ exchange the paired bits. Each reads only the opposite frozen bank and uses W as its separate dirty tile auxiliary; its weighted offset uses the target bank's own native place values. Every quotient label is fixed. The three banks use at most $3L/40$ address bits. After padding the one remaining suffix to 2^s and borrowing a dock of width $c \le L/40+1$, its low index supplies $R_{\mathrm{op}} = 2^{s-c}D \ge 2^{9L/10-O(1)}D$ opaque bits per record. This pays the weighted-mask metadata and exact boundary repair even when $D = 1$. There are no phase-role bits in these direct masks. The completed masks restore the dock and fix the low suffix index, so the suffix pad is cropped before the inverse layout.

Each mask costs $O(V(u \log R)^\tau + V)$. Since $\gamma > \tau$ and $h \log R \le L/40$, a batch costs $O(VL^\gamma)$. From $L - 2 < \log_2 M \le t \log_2 R$, $h \ge L/(80\ell_R)$ for sufficiently large admissible L. Together with $t \le L/(3 \log_2 L)$ this gives $O(\log R)$ batches. Summing proves the bound. Reversing the actual word gives the inverse; the bounded remaining range uses a direct finite branch. □

G.2 Permutations on exact even dimensions

An adjacent pair can select either I or the exchange matrix J. Over $\mathbb{F}_4$, both have a common word of exactly eight H letters interleaved with unit diagonals. Tensoring a slot uses the supplied canonical kernel. The diagonal coefficient factors by local axes; its lexicographic list is generated by prefix products in $O(M)$ fixed-field operations. The completed word is binary. Its lane pad can be removed only at that completed boundary.

Odd-start adjacent matchings are conjugates by coordinatewise $+1$. Explicitly, if $T(x) = x + 1 \bmod n$, an odd-start pair $(i, i+1)$ is TKT^{-1} with K switching $(i-1, i)$. Set $K = I$ on every irrelevant pair, in particular $(n-2, n-1)$ whose conjugate would wrap to $(n-1, 0)$. Thus the completed matching has no cross-boundary or wrap switch, even when its temporary K crosses a ragged parent boundary. At a synchronized odd-start stage, translate every axis in the fixed target set, including axes whose matching is identity. Choose $K = I$ on those axes; conjugation then cancels their translations. Thus the translated set, flat offset and source mask are independent of switch settings, including any control address in the controlled compiler. This selective translation costs $O(V)$: flat rotation by the sum of the selected native place values agrees except when a selected digit wraps. The exceptional fraction is at most $t/L^3 \le L^{-2}$. Rotate the good-source mask with the payload. For bad output slots, compute $x = P^{-1}(y)$ and the correct destination $T(x)$, sort those records by destination, and refill the holes. Since $P = T$ on good sources and both are permutations, $P(B) = T(B)$. This does not assume that the bad source set is invariant. Keys, record sorting and schoolbook decoding cost

$$O(\delta ML(D+L) + \delta ML^2) = O(V).$$

Here the L^2 exception-address bound does not perform t full L^2 divisions. If $b_i = \lceil \log_2 n_i \rceil$, then $\sum_i b_i \le 2L$. Streaming a current at-most-L-bit numerator through a b_i-bit remainder register divides by n_i in $O(Lb_i + L + b_i)$ tape steps. Summing gives $O(L^2)$ for all digits; the weighted Horner reconstruction has the same bound.

The source-good mask also has a linear implementation. Starting with one bit, append native factors in their physical order. For each bit a of the previous mask, print a for the $n_i - 1$ nonmaximal values and zero at the maximal value if axis i is selected; otherwise print n_i copies of a. Counter reset work is at most its ensuing n_i-bit output block. Since the intermediate mask lengths grow by factors at least two, their sum is below $2M$. Rotate that mask with the flat data permutation, so it marks exactly the output holes $P(B)$. Compute and copy a full key only at a bad record. A monotone output counter otherwise advances amortized in $O(M)$ total time. Stable binary radix sorting in L passes costs $O(KL(D + L))$ for $K \le \delta M$ bad records, including the full payload and header scans. In each pass stage one fixed-width item, remember its selected key bit, and copy the item to the corresponding stable zero/one bucket. Concatenate the buckets and rewind; a constant number of tapes and $O(K(D+L))$ steps suffice per pass, including staging and returns. Binary counters have amortized linear cost over each scan; their per-item resets cost at most that item's width. Since the sorted destinations are exactly $P(B) = T(B)$, refill bad holes in that order; there is no L-bit key comparison at each good position. This establishes the stated bound even for $D = 1$. The negative translation uses nonzero-digit source masks and the same proof.

Lemma 10 (Ragged global-shuffle network)**.** *Every permutation of an original even n has a word of $O(\log n)$ levels consisting of adjacent switches and the* same *full S_n and S_n^{-1}. No recursive odd-sized child is padded.*

Proof. At a level, the residual permutation preserves *each* interval setwise in a public ordered partition into contiguous parents. Pair positions from each parent's start, leaving its last position unmatched if its size is odd. Use the same output pairing. The bipartite multigraph has input pair/singleton vertices, output pair/singleton vertices, and one edge per packet. Even parents give even cycles. An odd parent gives one path between its two singleton vertices, plus even cycles. That path has an odd number of edges. Its endpoint edge colors can both be forced to the global parity of the parent's last position.

Alternating edge colors give input switches E_{in} and output switches E_{out} so that the residual has identity

$$\pi = E_{\text{out}} S_n C S_n^{-1} E_{\text{in}}.$$

The full inverse shuffle stably splits global even and odd positions. For original even n, write this split as

$$p_n(x) = S_n^{-1}(x) = \begin{cases} x/2, & x \text{ even}, \\ n/2 + (x-1)/2, & x \text{ odd}. \end{cases}$$

A parent interval $[a, b)$ therefore has the two image fragments

$$[\lceil a/2 \rceil , \lceil b/2 \rceil), \qquad [n/2 + \lfloor a/2 \rfloor , n/2 + \lfloor b/2 \rfloor).$$

The even fragments concatenate to $[0, n/2)$ and the odd fragments to $[n/2, n)$, skipping empty fragments. The new contiguous block list is

$$\text{[all parents' even fragments][all parents' odd fragments].}$$

The endpoints depend only on n and the parent partition, independently of π and its switch colors. The residual C preserves each new fragment setwise. The recursive call is on this single n-wire residual with the whole updated partition. Thus each level adds only a fixed number of slot types to one recursive word. Track this order; do not gather siblings. Each fragment has

size at most the ceiling of half its parent. After $\lceil \log_2 n \rceil$ levels all blocks are singletons. Each adjacent switch stage splits into its even-start and odd-start families. Short axes can continue passive singleton levels; their remaining shuffles cancel on unwind. □

Synchronize the networks for all axes to $\lceil \log_2 R \rceil$ levels. Lemmas 9–10 and the adjacent-switch words give a generic local-permutation tensor cost

$$O(VL^{\gamma} \operatorname{polylog} L). \tag{33}$$

This retains the parameter of the supplied kernels: each adjacent-switch word uses a fixed number of native-H calls, each shuffle has the same γ tariff, and there are $O(\log R)$ levels. Translations and fixed-field diagonal scans cost $O(V)$. Thus synchronization adds only a fixed power of $\log L$. Public local tables and network settings have polynomial-time sequential generation in R, L.

G.3 Controlled tensors and the control gauge

Partition axes into targets T and controls C. For each control address c let $\pi_i(c)$ be a permutation on target axis i. Controls are address coordinates, not input-dependent payload values.

Lemma 11 (Controlled compiler)**.** *Let $G(R, L) \geq 1$ bound the time to generate one outer $\mathbb{F}_4$ scalar coefficient at a current record address, including address decoding, control-gauge inversion, all required local tables and ragged settings. Assume this is a fixed polynomial-time sequential procedure with fixed tapes. Then the controlled map*

$$(c, x_T) \longmapsto (c, (\pi_i(c)(x_i))_{i \in T})$$

has bit cost

$$O\big((MDL^{\gamma} + MG(R, L)) \operatorname{polylog} L\big)$$

with fixed tapes and ordinary order. In particular $G(R, L) = O(D)$ gives (33). For the explicit CRT families below, $G(R, L) = O((R + L + 2)^{12})$; thus $D \geq (R + L + 2)^{64}$ suffices.

Proof. First construct the completed subset kernel $K_T = H_T \otimes I_C$. Use a static common eight-H word for H versus I, tensoring each inner letter across all axes. The control axes may move *inside* this implementation. At its completed boundary, K_T fixes them. Insert controlled scalar stages only between completed K_T calls. The outer eight-H switch word is then valid on each fixed-control fiber. Its coefficient is $\kappa(x, c) = \prod_{i \in T} \kappa_i(x_i, c) \in \mathbb{F}_4^*$. We do not assume factorization by control axes.

The fragment-endpoint formulas in Lemma 10 make the partition and shuffle topology independent of the control address. Only the switch settings depend on c, so a single fixed slot schedule applies to all fibers. Use the full tensor shuffle on *all* axes. Give C identity switches. At an outer slot their temporary physical value is $q_e(c) = S_C^e(c)$, where e is the signed number of completed forward minus inverse shuffles. At each scalar stage enumerate the current native lexicographic addresses and decode logical $c = q_e^{-1}(x_C)$. Generate the target settings for this logical c. At every odd-start stage use the same translation of all axes in the fixed set T, with identity $K_i(c)$ on idle axes or fibers. The flat approximation can temporarily carry into a control coordinate on bad records. Complete its exact repair before reading any control-dependent setting. Completed subset kernels and translations then preserve the physical controls; a shuffle increments e in its prescribed direction. Induction over the word proves the fiber invariant. Descent and unwind balance, so $e = 0$ at exit and the controls are restored.

For the CRT affine families, one concrete generator decodes addresses by schoolbook division, computes the control CRT sums, generates the at-most-R local table, and constructs the pair graph by sequential scans. All axes, levels, records, and full-list lookups are bounded in the following generator analysis by $O((R + L + 2)^{12})$ time. Fix the dominating polynomial $P_1 =$

$(R+L+2)^{64}$ before choosing D. For each record park its D bits locally, generate one coefficient, apply its fixed two-bit map, and advance. In general this scalar stage costs $O(M(D+G))$; visited work intervals and returns are included in the generator bound. There are only polylog L outer stages, proving the displayed general bound. The CRT generator just described has $G = O((R+L+2)^{12})$, which the chosen word width dominates, so its scalar stage costs $O(V)$. No whole-array scan is nested inside a per-record lookup. This also aligns coefficients to the actual post-shuffle stream order. □

Lemma 12 (Sequential control-coefficient generation)**.** *For the affine CRT permutations used below, a coefficient at one current native address, at one specified network slot, can be generated on a fixed number of tapes in $O((R+L+2)^{12})$ bit time, including decoding, work initialization, erasure and head returns.*

Proof. Let $T = R+L+2$. All integers representing a native address or a CRT product have $O(L)$ bits. Schoolbook division, multiplication and extended Euclidean arithmetic have fixed polynomial cost in that width. Decode the current mixed-radix address, invert the signed shuffle gauge, and reconstruct any control block x_A by ordinary CRT. There are at most L axes, so repeated schoolbook operations and sequential scans give the deliberately loose bound $O(T^6)$ for this step.

For each target axis, list the at most R images of the affine permutation $z \mapsto P_A z + x_A \bmod p$, fixing its dummy value. The ragged network above constructs its settings from these explicit lists. There are at most R^2 fragment and pair records even if every recursion level is stored separately. A record contains $O(L)$ bits. Determine each local switch by tracing the alternating pair graph, always looking up an endpoint by a full sequential scan. Fewer than $O(R^2)$ path steps per fragment and $O(R^2L)$ bits per lookup suffice. Allowing a separate traversal for every fragment and every one of at most L axes costs $O(R^6L^2)$, which is $O(T^8)$. Endpoint arithmetic and the fixed common eight-H word then specify the requested scalar. Multiply the at most L nonzero $\mathbb{F}_4$ entries in its tensor list.

These are nested loops of the stated finite bounds. A list lookup, scratch initialization, return and erasure each has a complete scan already charged above; replacing any direct table access by such a scan still fits $O(T^{12})$. The generator receives only the public address and slot descriptor, and performs no scan of the payload array. The bound is therefore a per-record bit cost G, giving MG for M records, rather than MDG for D payload bits per record. □

G.4 Enlarging the cyclotomic word

Lemma 13 (Uniform host initialization)**.** *From the binary input length n, for all sufficiently large n a deterministic fixed-tape procedure constructs a 2-admissible tuple $(q_0, \dots, q_e)$, its product $2n < N < 4n$, the attached common-multiple data, and $\alpha, \Phi_\alpha, \rho$ with $q_0 \mid \alpha$ and ρ a principal (N/q_0)-root. Its bit time is $N^{o(1)}$, including writing and verifying the factor and root data.*

Proof. Apply [1, Proposition 3.4] at $p = 2$ and target $2n$; its tuple and auxiliary data are part of the constructed admissible length. Apply Proposition 3.8 of that source to this tuple. Their respective time bounds are $2^{O((\log\log n)^2)}$ and $2^{O((\log\log N)^4)}$ at fixed $p = 2$. Both are $N^{o(1)}$ in the source's deterministic fixed-tape model. Write $L_0 = \lceil \log_2 N \rceil$ and $\ell = \lceil \log_2 L_0 \rceil$; its divisor has $\log \alpha = O(\ell^4)$.

Verification uses only polynomial-size setup operations. Trial division checks the emitted prime list and its product, since each prime is at most $R_0 = 2^{\ell^2}$; monic polynomial arithmetic checks the generated cyclotomic data. Put $K = N/q_0$. Check $\rho^K = 1$ and

$$\gcd(\rho^{K/q_i} - 1, \Phi_\alpha) = 1 \qquad (1 \le i \le e).$$

The squarefree quotient is a product of fields. These tests force order K in every component, hence principality by the geometric-sum identity. Modular powers, gcds and all certificates cost

a polynomial in α, R_0, L_0, which is $2^{O(\ell^4)} = N^{o(1)}$. Schoolbook arithmetic suffices in this setup; it uses no call to the proposed multiplier. □

If n is implicit in end-marked coefficient lists, counting and copying them costs $O(n)$ by amortized counters. The bounded initial length range uses the direct branch; no factorization is supplied as advice.

Use the unconditional host of [1]. Put $L_0 = \lceil \log_2 N \rceil$, $\ell = \lceil \log_2 L_0 \rceil$ and $R_0 = 2^{\ell^2}$. Its squarefree N has prime factors in $(L_0^3, R_0]$. The constructed divisor α uses $O(\ell^2)$ factors and satisfies

$$\log \alpha = O(\ell^4), \qquad -\log \frac{\varphi(\alpha)}{\alpha} = O(\ell^2/L_0^3).$$

The distinguished prime q_0 divides α. Proposition 3.8 constructs ρ, a principal (N/q_0)-root in $\mathbb{F}_2[Y]/\Phi_\alpha(Y)$, as well as this cyclotomic polynomial. Then $\omega = \rho^{\alpha/q_0}$ is a principal $m = N/\alpha$ root. Indeed for $s = \alpha/q_0$ and $0 < j < m$, the old $K = sm$ character sum at exponent sj is s copies of the m-sum. Odd s is one in characteristic two, so that sum vanishes.

Append unused prime factors until $\alpha' = \alpha J$ satisfies

$$D' = \varphi(\alpha') \geq (R_0 + L_0 + 5)^{64}.$$

Only $O(\ell)$ factors are needed because each multiplies D by more than L_0^2. There are $\Omega(L_0/\ell^2)$ available factors. The first-crossing choice implies

$$\log \alpha' = O(\ell^4), \qquad \varphi(\alpha')/\alpha' > 1 - 1/L_0, \quad m'D' = \Theta(N), \quad m' = N/\alpha'.$$

We do not claim the enlarged divisor meets the original definition's literal numerical upper cutoff; these displayed estimates are sufficient. We retain the stronger density estimate as well. The original divisor has $O(\ell^2)$ prime factors, and enlargement adds only $O(\ell)$, all greater than L_0^3. Thus

$$\theta := 1 - \frac{D'}{\alpha'} = 1 - \prod_{p \mid \alpha'} (1 - p^{-1}) \leq \sum_{p \mid \alpha'} p^{-1} = O(\ell^2/L_0^3). \tag{34}$$

This follows from the actual divisor construction in [1, Proposition 3.8, proof], rather than just its weaker admissible-divisor definition.

The substitution $Y \mapsto U^J$ defines a unital homomorphism

$$\mathbb{F}_2[Y]/\Phi_\alpha(Y) \longrightarrow S = \mathbb{F}_2[U]/\Phi_{\alpha'}(U),$$

because $\Phi_{\alpha'}(U)$ divides $\Phi_\alpha(U^J)$. Write ι for this homomorphism. It preserves principal-root character sums. The required root is $\omega' = \iota(\omega)^J$: it satisfies $(\omega')^{m'} = 1$. For $0 < j < m'$, split the old $m = Jm'$ terms into J equal periods to obtain

$$0 = \sum_{a=0}^{m-1} \iota(\omega)^{Jja} = J \sum_{b=0}^{m'-1} (\omega')^{jb}.$$

Odd J is one in characteristic two, proving principality. Thus no new root oracle is needed.

Pad the remaining prime axes by $n_i = p_i + 1$. Then $M/m' = \prod_i (1 + 1/p_i) = 1 + O(L_0^{-2})$. Since $\alpha' > L_0^{64}$, for sufficiently large N, $L = \lceil \log_2(2M) \rceil \leq L_0$ and $L = \Theta(L_0)$. The native radix hypotheses and $D' \geq P_1(R_0 + 1, L)$ hold together. Setup polynomial in R_0, D', L_0 is $N^{o(1)}$ and is charged once.

G.5 The coupled CRT index permutation

Write $m' = \prod_{i=1}^{t} p_i$, with least-significant-first digits

$$x = a_1 + p_1 a_2 + \cdots + (p_1 \cdots p_{t-1}) a_t.$$

Its physical lexicographic axis order is $[p_t] \cdots [p_1]$, already ordinary integer index order. In one lexicographic scan embed these records in the $[p_i + 1]$ grid, preserving their order, and zero-fill every address containing at least one dummy coordinate.

Lemma 14 (Parallel CRT tree)**.** *The permutation $a \mapsto (x \bmod p_i)_i$ and its inverse have cost $O(m' D' L_0^{\gamma} \operatorname{polylog} L_0)$.*

Proof. For a contiguous block $I = A \cup B$ with A lower significance, write $x_I = x_A + P_A x_B$, $P_A = \prod_{i \in A} p_i$. After converting both children to their local residues, the left residues are already correct for x_I. Each right target becomes

$$z_i \longmapsto P_A z_i + x_A \pmod{p_i}.$$

This is a permutation because P_A is a unit modulo p_i. The left residue address determines $x_A \in [0, P_A)$ by ordinary integer CRT arithmetic. Fix the target dummy, and use the identity when a relevant left control is dummy. This is an explicit bijection on every fiber and preserves every dummy-pattern class at completion.

At each level of a balanced tree take the union of all right axes as T, and all left/passive axes as C. The disjoint node merges are one controlled tensor from Lemma 11. There are $O(\log(t+1))$ levels. This schedule avoids two separate full-volume child executions per node. Reverse the stages and affine maps for the inverse. □

G.6 Characteristic-two local Fourier compilation

For an odd prime p, let $u \in S$ be a principal p-root. Every field component has exact order p, so $1 - u^j$ is a unit for $0 < j < p$. Set $H_j = \prod_{a=1}^{j} (1 - u^a)$ and $H_0 = 1$. The Newton matrix

$$N_{j,k} = \prod_{a=0}^{k-1} (u^j - u^a), \quad j \geq k,$$

is diagonal–lower-Toeplitz–diagonal, with $N_{j,k} = H_j H_{j-k}^{-1} (-1)^k u^{k(k-1)/2}$. The symmetric Newton factorization is $F_p(u) = N \operatorname{diag}(N_{kk})^{-1} N^T$. All pivots remain units over the finite etale algebra.

The explicit formal triadic circuit in Lemma 6, with one operand fixed, supplies a linear convolution DAG of size $O(b \log^8 b)$ and depth $O(\log^2 b)$ over every commutative $\mathbb{F}_2$-algebra. Bounded-degree copy trees are included. Appendix F also gives explicit constant/linear gate rules for this formal specialization, with no Boolean idempotence identity substituted in the coefficient ring.

The borrowed-coordinate mechanism of [5] restores arbitrary dirty workspace by replaying once with the true sources and once with the sources suppressed. Its algebra works in characteristic two. Inside a width-v Toeplitz recursion, choose chunk size $b = cv/\log^8 v$, with c a sufficiently small fixed constant. The two chunks and all gate coordinates fit within the same v positions. There are $O(\log^{16} v)$ source/target chunk pairs, each using $O(\log^2 v)$ replay rounds. A balanced recursion gives $O(\log^{19} p)$ rounds on exactly p coordinates, as proved in Lemma 7. Its source, target and borrowed ports are disjoint; output-copy wires are counted inside the DAG. All borrowed values return before the next chunk pair. The finite small range is direct.

Work in $B = S \otimes_{\mathbb{F}_2} \mathbb{F}_4$ and take central $\eta^2 + \eta + 1 = 0$. Every coefficient $\mu \in B$ is the sum of two units: let e be one on the components where $\mu = 1$ and zero elsewhere, and put

$d = 1 + e(\eta + 1)$. Then d and $\mu + d$ are units. All component degrees equal $r_B = \mathrm{lcm}(\mathrm{ord}_{\alpha'} 2, 2)$; thus $e = 1 + (\mu + 1)^{2^{r_B}-1}$ can be prepared by polynomially many small-algebra operations. For unit t,

$$U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} = \mathrm{diag}(t, 1) H J \, \mathrm{diag}(t^{-1}, 1), \qquad U(\mu) = U(d) U(\mu + d).$$

Here is one uniform slot word for the whole round. Pad each p axis by its one dummy and complete a partial shear matching by pairing its unused coordinates with coefficient $\mu = 0$. Order every used pair as (target,source), and send these pairs to canonical adjacent pairs by a local permutation. Set $u_0 = d$, $u_1 = \mu + d$ and $D(t) = \mathrm{diag}(t, 1)$. In rightmost-first matrix order,

$$U(\mu) = D(u_0) H J D(u_0^{-1} u_1) H J D(u_1^{-1}).$$

All diagonal entries are units. An unused pair has $u_0 = u_1 = 1$, so its completed word is $(HJ)^2 = I$; it need not be fixed at an intermediate letter. Tensoring uses exactly two full canonical H slots, two full J slots and three axis-factorized diagonal slots. The pair relabeling and its inverse each cost one generic tensor permutation from (33); charge the two J slots there as well. Thus a round has two native-H calls and at most four generic permutation calls. No separate selection of H versus I is needed on this local path. Each completed local axis map is $F_{p_i} \oplus [1]$ on its native-valid range $n_i = p_i + 1$. The tensor preserves each per-axis dummy-pattern subspace; it need not fix every full address containing a dummy, since other axes still transform. Consequently an initially zero-filled dummy complement stays zero at the completed tensor boundary. Keep the full padded grid through these operations, including every intermediate letter.

Lemma 15 (Independent runtime word arithmetic)**.** *There is a uniform fixed-tape algorithm for ordinary binary polynomial products at length v in $O(M_{\mathrm{old}}(v))$ time, with bound function*

$$M_{\mathrm{old}}(v) := v \log^2(2v) \log\log(4v).$$

It uses classical integer multiplication, independently of the proposed main construction and of the formal convolution circuit supplier.

Proof. For $v \geq 2$ set $k = \lceil \log_2(v + 1) \rceil$ and lift $\sum_{i<v} a_i X^i$ to $\sum_{i<v} a_i 2^{ki}$, with $a_i \in \{0, 1\}$. Every integer convolution coefficient is at most $v < 2^k$. Consequently the product's k-bit chunks are the exact integer coefficients, without carries between chunks. Their low bits give the ordinary $\mathbb{F}_2$ product. The encoding, zero padding, chunk extraction and optional bit-order reversals are sequential scans costing $O(vk)$. Counter resets are charged to their ensuing chunks. The classical integer algorithm of [2] takes $O(B \log(2B) \log\log(4B))$ steps on a fixed multitape machine for $B = vk$ bits, proving the bound. Marking visited work-tape intervals allows head returns and cleanup within the same order of time. Length one is direct. The bound function is stable under fixed-factor length changes. □

Tensor scalar coefficients are prepared by prefix expansion in the current native order, as detailed in Appendix F. The total number of products is below $2M$, not one product per axis at every record. A B symbol has $2D'$ bits. Each B multiplication or fixed-scalar application costs $O(M_{\mathrm{old}}(2D')) = O(M_{\mathrm{old}}(D')) = D' \, \mathrm{polylog}\, L_0$ bit steps using Lemma 15 and the explicit fixed-modulus reduction in Appendix F. In contrast, the routing compiler's $\mathbb{F}_4$ scalars are fixed two-bit maps costing $O(D')$. These two costs must not be conflated. Project back to S only after the complete Fourier tensor has returned to that subspace. Each synchronized replay round has a fixed number of calls at the generic/native γ tariff and a fixed number of these scalar stages. There are $O(\log^{19} R) = \mathrm{polylog}\, L_0$ rounds, so the local Fourier tensor retains the same γ exponent.

G.7 Ordinary multiplication and the time ledger

Use Lemma 13 to construct $N = \Theta(n) > 2n - 2$ and its certified factor list, then enlarge the divisor as above. Initial input scans cost $O(n)$ and all host initialization is included in the $N^{o(1)}$ setup term below. The top Agarwal–Cooley isomorphism from [1, Lemma 2.2] is

$$X \longmapsto Y^c Z, \quad c = (m')^{-1} \bmod \alpha',$$

from $\mathbb{F}_2[X]/(X^N - 1)$ to $\mathbb{F}_2[Y, Z]/(Y^{\alpha'} - 1, Z^{m'} - 1)$. Its coprime factors α', m' both exceed two for sufficiently large n: α' tends to infinity and $\log \alpha' = o(\log N)$. It uses one classical rectangle transpose and cyclic row shifts in $O(N \log \alpha') = N \operatorname{polylog} L_0$ bit time, in either direction. Explicitly, an exponent $k = r + m'j$ goes to (Z-index, Y-index) $= (r, cr + j \bmod \alpha')$; this fixes the row-shift convention and its inverse. Since α' is odd, $\Phi_{\alpha'}$ is coprime to $\psi = (Y^{\alpha'} - 1)/\Phi_{\alpha'}$. Put $f = \Phi_{\alpha'}$ and $d = \deg \psi = \alpha' - D'$. If $d = 0$ the coefficient split is the identity. Otherwise use Lemma 8 on each α'-bit row for both projections, including the possibly much shorter modulus ψ. The latter's prepared reciprocal has length $\alpha' - d = D'$. Choose fixed Bezout polynomials $uf + v\psi = 1$, with $\deg u < d$ and $\deg v < D'$. The idempotents $e_f = v\psi$ and $e_\psi = uf$ have degree $< \alpha'$, and reconstruct

$$A = (A_f e_f + A_\psi e_\psi) \bmod (Y^{\alpha'} - 1).$$

Two padded length-α' word products and a linear cyclic fold suffice for this reconstruction. Projections and reconstruction thus cost $O(M_{\text{old}}(\alpha'))$ per row, and

$$O(m' M_{\text{old}}(\alpha')) = N \operatorname{polylog} L_0$$

overall. All reciprocals, Bezout coefficients and idempotents are public polynomial-α' setup, charged once. The two projected streams have total length $m'(D' + d) = N$ bits.

For the large component let ω be the principal m' root and

$$E_i = (m'/p_i)((m'/p_i)^{-1} \bmod p_i), \quad \omega_i = \omega^{E_i}.$$

The CRT idempotents give $\omega^{xy} = \prod_i \omega_i^{(x \bmod p_i)(y \bmod p_i)}$. Lemma 14 and the local Fourier tensor therefore compute the true ordinary cyclic DFT. Two forward transforms, point products and one inverse compute the product in $S[Z]/(Z^{m'} - 1)$. Retain the full native grid through these completed operations; its dummy complement remains zero at each such boundary and under the point product, so it can be cropped after inverse routing. The inverse uses inverse roots and reversed index routing; its odd normalization is one in characteristic two.

Here the large coefficient ring obtained by projection is $\mathbb{F}_2[Y]/\Phi_{\alpha'}(Y)$, identified with S by renaming Y to U. This identification is distinct from the earlier setup homomorphism $Y_{\text{old}} \mapsto U^J$ from the smaller ring $\mathbb{F}_2[Y_{\text{old}}]/\Phi_\alpha(Y_{\text{old}})$. That homomorphism constructs the root; it is not applied a second time to the new input words. In this section ω denotes the already constructed ω'. Indeed if $J > 1$, U^J has order α, rather than α', in every field component of S. Hence $\Phi_{\alpha'}(U^J)$ is a unit, so that repeated substitution would not even descend from the new quotient.

Frequency order needs its own verification. If local roots instead have exponents $b_i h_i$, with $b_i = m'/p_i$ and h_i a unit modulo p_i, an output digit s_i describes frequency residue $b_i h_i s_i$. The idempotent choice $h_i = b_i^{-1}$ makes that residue s_i. For a coordinate permutation Q, however, $A = QF$ still satisfies $A^{-1}((Aa) \odot (Ab)) = F^{-1}((Fa) \odot (Fb))$. Thus matching inverse and product checks alone do not certify a literal Fourier index convention.

The complement has $d = \deg \psi = \alpha' - D'$ and $v = dm' = N\theta = O(N\ell^2/L_0^3)$ by (34). If $d = 0$, omit this component. Pack $Y^j Z^i \mapsto T^{2di+j}$ in $\mathbb{F}_2[T]$. There is no overlap between coefficient-product slots because $j + j' < 2d$. Each input has degree at most $2v - d - 1$ and the product degree at most $4v - 2d - 2$. Use Lemma 15 at input length $2v$. Crop its known zero tail to

$2d(2m'-1)$ coefficient slots and unpack $2m'-1$ rows of $2d$ bits, each with final bit zero. Reduce their degree-$\leq 2d-2$ polynomials modulo the fixed monic ψ using Lemma 8, then fold $Z^{m'}=1$. All packing, unpacking and folding scans cost $O(v)$; when $d=1$ the row remainders are direct. The total cost is $O(M_{\mathrm{old}}(2v)+m'M_{\mathrm{old}}(d))=O(M_{\mathrm{old}}(v)+m'M_{\mathrm{old}}(d))$. Since $d\leq v\leq N$, each logarithmic factor is bounded using L_0, so the total is

$$O(vL_0^2\ell)=O(N\ell^3/L_0)=o(N).$$

The scans and all reductions are included. The weaker estimate $v<N/L_0$ alone would not pay for this integer-packing fallback; the stronger density is essential here. No integer packing of the full ordinary input is used.

Lemma 16 (Typed top-level execution)**.** *The preceding component algorithms compose on a fixed number of streams, with ordinary coefficient input and output. Their complete calls use the stated tariffs; no coefficient projection or outer native-dummy crop occurs inside a Fourier word.*

Proof. Use the following order, with a, b denoting the two inputs.

1. Count and copy the coefficient lists, construct the host and all length-dependent tables, and zero-fill both lists to N coefficients. The live top-level input streams contain $2N$ bits. Their cost is $O(n)+N^{o(1)}$ before the linear zero-fill.

2. Apply the top isomorphism to each list, producing m' consecutive α'-bit coefficient rows. Project each row to the f and ψ components. The two large streams contain m' words of D' bits; the two small streams contain m' words of d bits. Park the small streams on a fixed separate pair of tapes. The maps and projections cost N polylog L_0. The large words are identified with S by the variable rename just specified.

3. Embed each large word in $B=S[\eta]/(\eta^2+\eta+1)$ as $a+0\eta$, and zero-fill its native dummy grid. Keep these two M-word B arrays through both forward transforms, their point product and the inverse. They occupy $4MD'=O(N)$ bits at these top-level boundaries. Transform one while the other waits on its own tape; constant whole-array passing and parking scans cost $O(N)$. No callee traverses the parked array.

4. To specify every completed transform, let $E: B^{m'}\to B^M$ be the zero-fill embedding from integer mixed-radix order. Let Q be the completed extended CRT permutation, and put

$$\mathcal{T}_\omega=\bigotimes_i(F_{p_i}(\omega^{E_i})\oplus[1]),\qquad \mathcal{D}_\omega=Q^{-1}\mathcal{T}_\omega Q.$$

   The CRT exponent identity gives

$$\mathcal{D}_\omega E=EF_{m'}(\omega). \tag{35}$$

   Thus the output is again in ordinary integer frequency order, in the same native grid. The whole operators Q and $\mathcal{T}_\omega$ preserve every dummy-pattern subspace. Their intermediate words may mix the auxiliary coefficient lanes or dummy values; only completed boundaries are used in this identity. Their valid block has entries in S, so a complete transform returns S-valued words when its input was S-valued. Keep the unused η lane allocated nonetheless.

5. Replace the first completed transform array by the coordinatewise B product of the two arrays, scanning them together, and erase the second array. On valid coordinates this is exactly the S product; on every dummy coordinate it is zero. Performing all M products, without an exceptional-address lookup, costs $O(MM_{\mathrm{old}}(D'))=N$ polylog L_0. Apply $\mathcal{D}_{\omega^{-1}}$ to the first array. The principal-root character sums imply $F_{m'}(\omega^{-1})F_{m'}(\omega)=m'I=I$.

Equation (35) and the convolution identity therefore give the cyclic product in $S[Z]/(Z^{m'}-1)$. Explicitly, for the projected coefficient vectors a_f, b_f the jth inverse coefficient is

$$\sum_{r,s=0}^{m'-1}(a_f)_r(b_f)_s\sum_{k=0}^{m'-1}\omega^{k(r+s-j)}=\sum_{r+s\equiv j \pmod{m'}}(a_f)_r(b_f)_s.$$

The inner character sum is zero off the stated congruence and is $m'=1$ on it. Thus the complete retained-grid computation is $\mathcal{D}_{\omega^{-1}}((\mathcal{D}_\omega Ea_f)\odot(\mathcal{D}_\omega Eb_f))=E(a_f *_{\mathrm{cyc}} b_f)$. Only now remove the zero η lane and crop the zero dummy complement, using monotone native-address scans costing $O(N)$. The result is m' consecutive D'-bit words in ordinary Z order. Generate the dummy test once per record using the fixed degree-twelve address procedure; its cost is paid by $D'\geq P_1$.

6. Compute the parked small-component product by the independent packing and fixed-monic reductions above, at cost $o(N)$, obtaining m' consecutive d-bit words, omitting this branch if $d=0$. Reconstruct the α'-bit rows from the two components and apply the inverse top isomorphism at cost $N\operatorname{polylog} L_0$. Extract the first $2n-1$ coefficients in a final $O(N)$ scan. All other top-level streams are erased and their heads returned by scans charged to their lengths.

There are three complete transforms, each with a constant number of CRT maps and the stated synchronized local rounds. Their total is $O(NL_0^\gamma \operatorname{polylog} L_0)$, since their B payload width is $2D'$ and hence satisfies the same metadata lower bound. The number of top-level streams is fixed; setup archives and recursive scratch use the separately charged tape families, rather than one fresh tape per axis, table or recursion level. □

The full-volume routing and Fourier arrays have $V=\Theta(N)$ and address width $L=\Theta(L_0)$. The smaller complement arrays have the separate budget above. The stage counts are fixed powers of $\log L_0$. The complete bound furnished by the construction is

$$O(NL_0^\gamma(\log L_0)^C)+N\operatorname{polylog} L_0+N^{o(1)}.$$

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